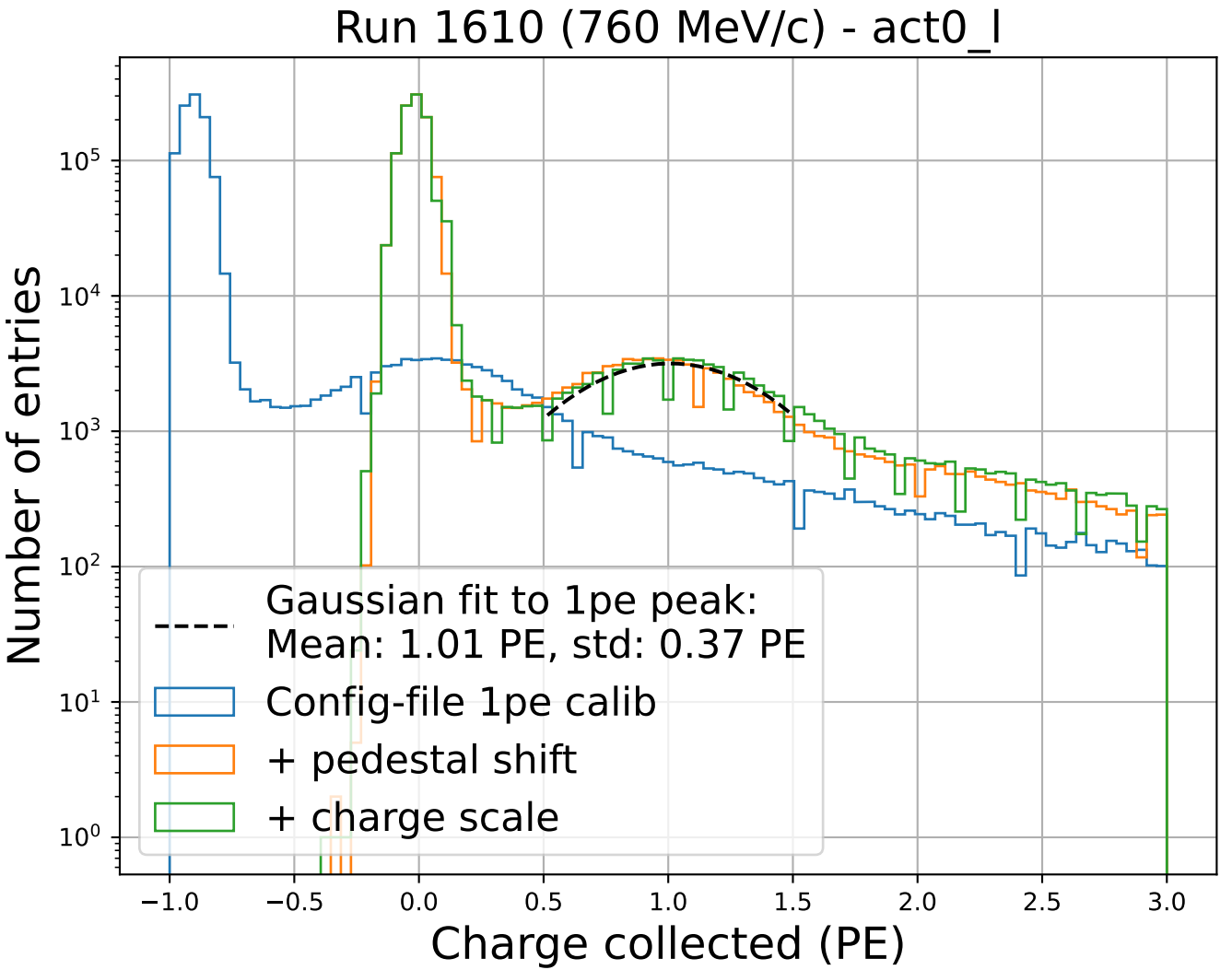
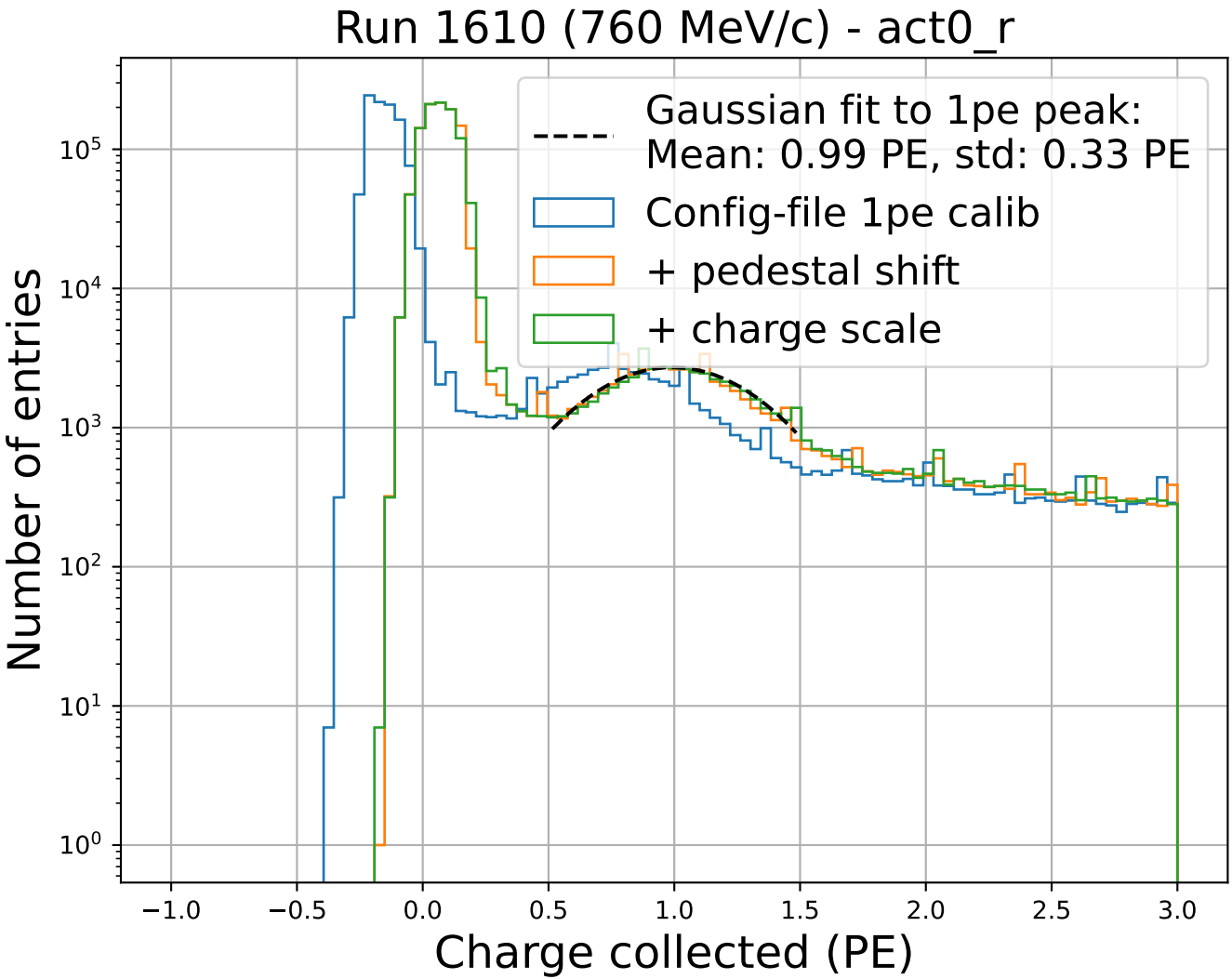
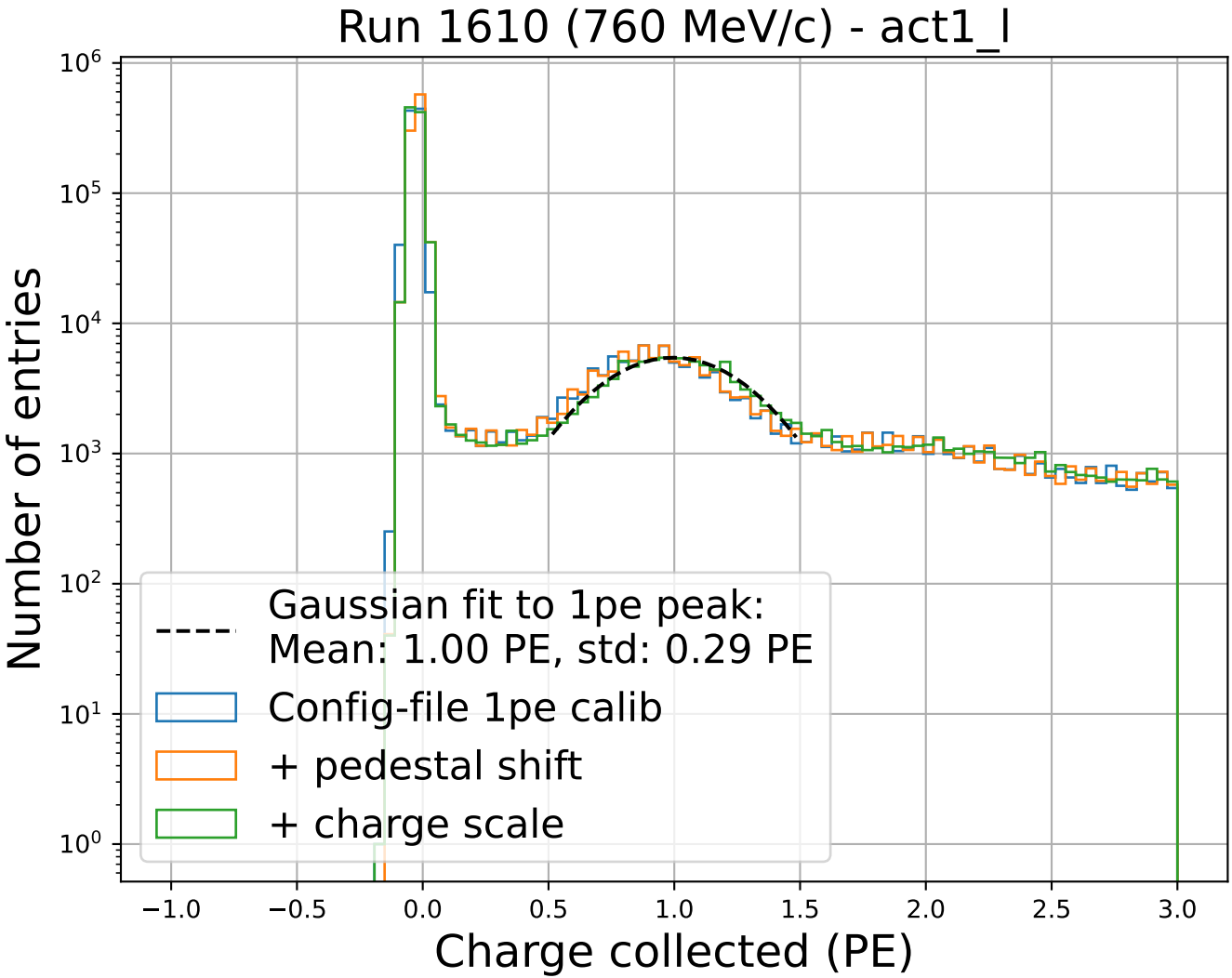
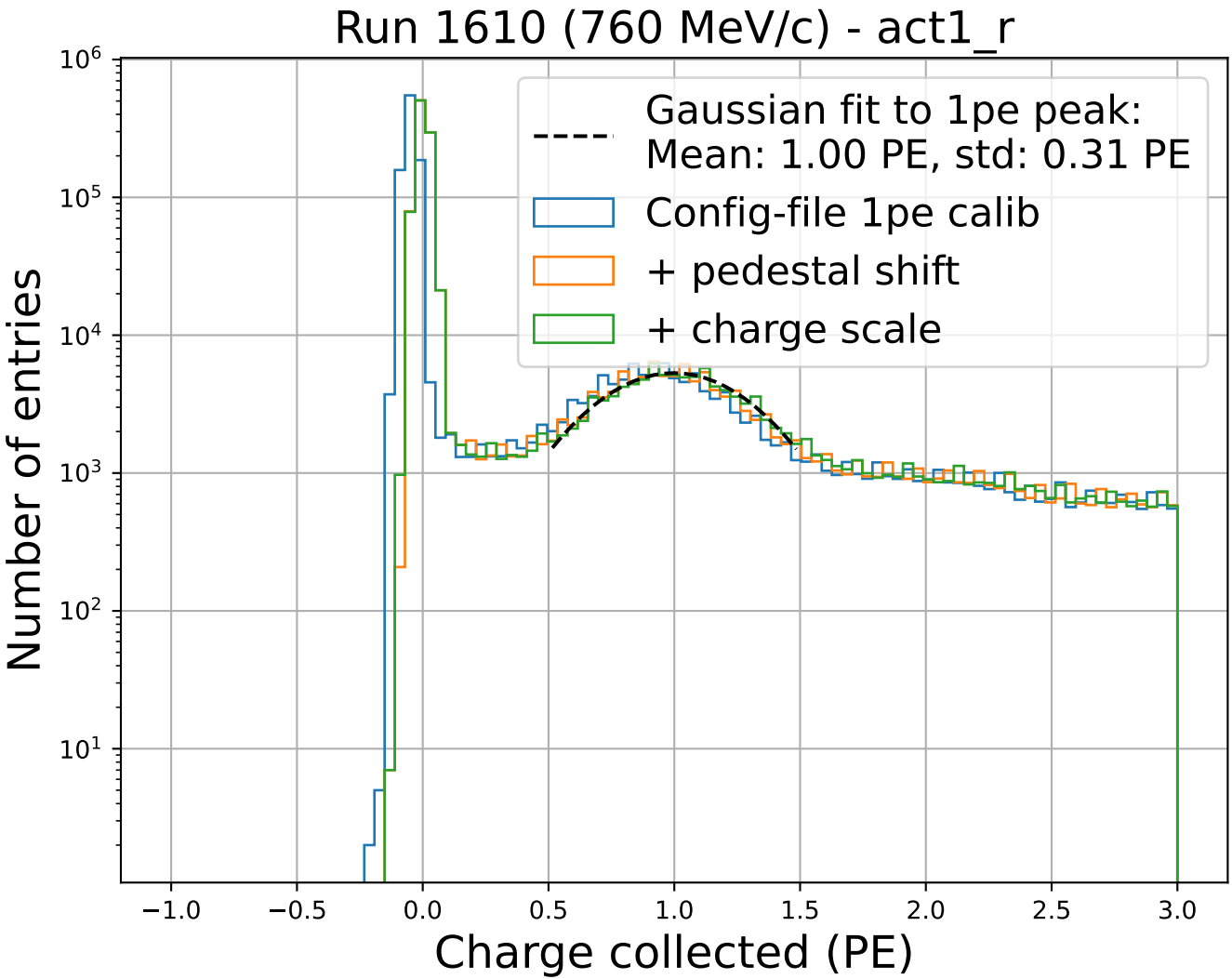
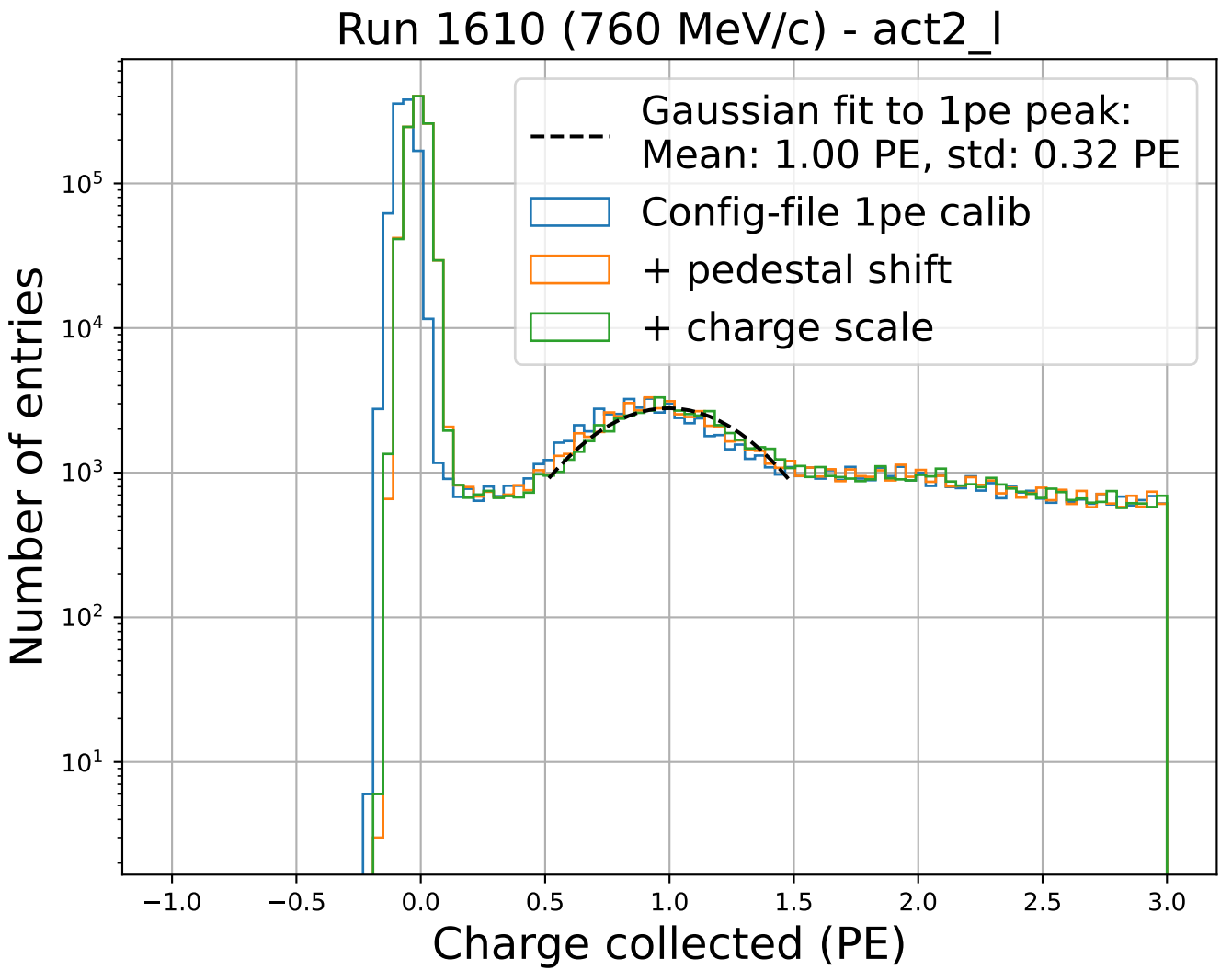
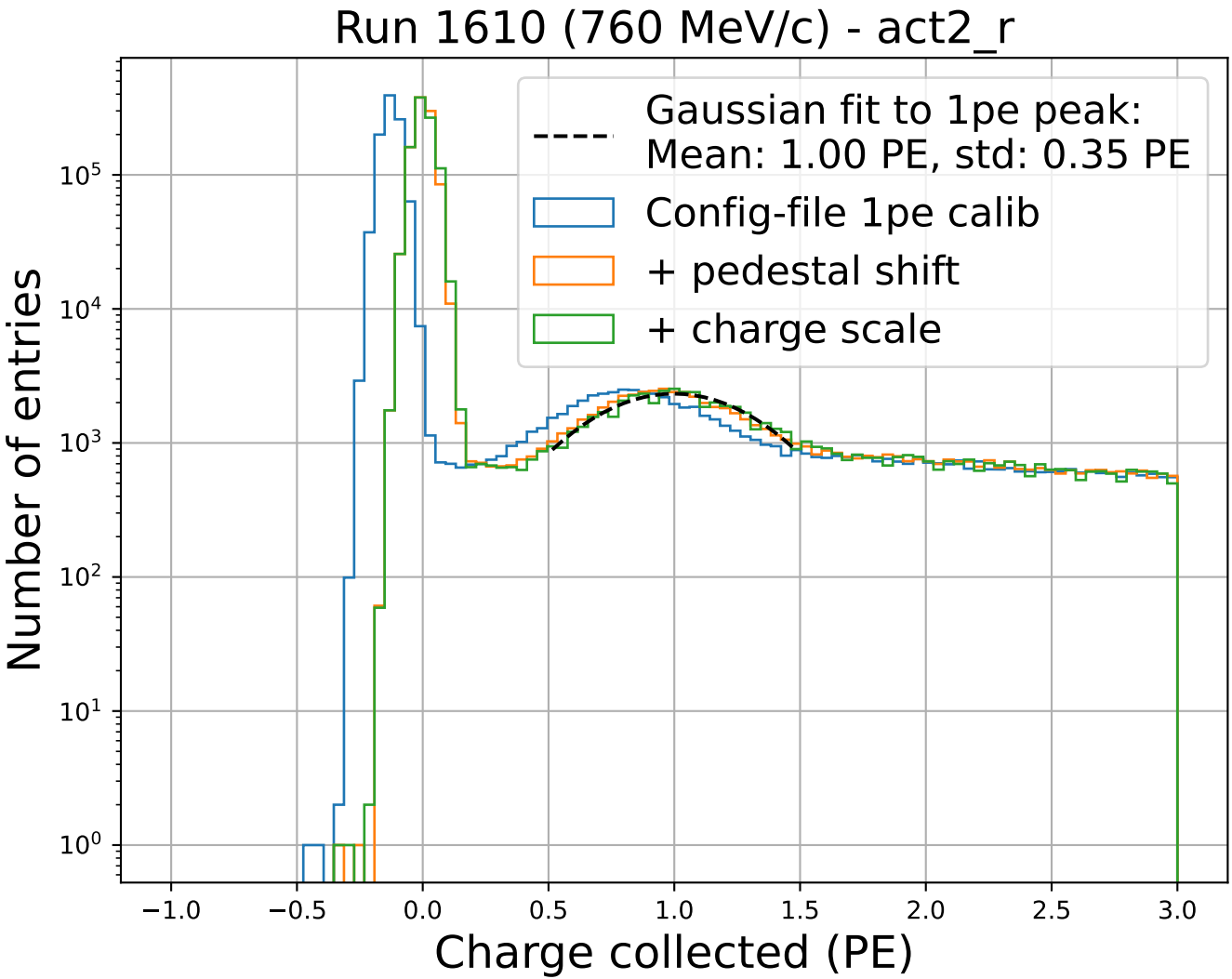

Run 1610 (760 MeV/c) - act0_l
Number of entries
Charge collected (PE)
Gaussian fit to 1pe peak:
Mean: 1.01 PE, std: 0.37 PE
Config-file 1pe calib
+ pedestal shift
+ charge scale
10^0
10^1
10^2
10^3
10^4
10^5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0

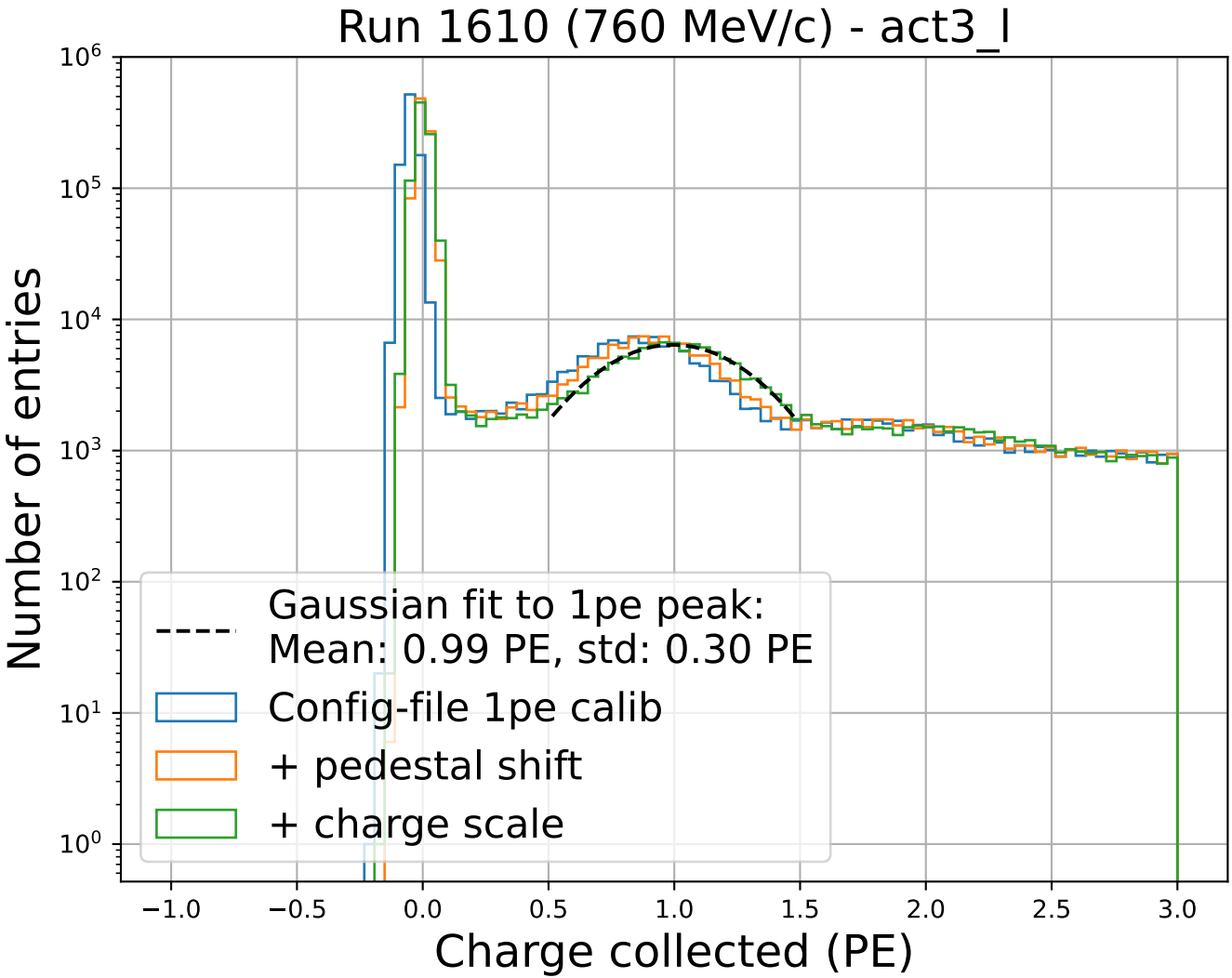

Run 1610 (760 MeV/c) - act0_r
Gaussian fit to 1pe peak:
Mean: 0.99 PE, std: 0.33 PE
Config-file 1pe calib
+ pedestal shift
+ charge scale
Number of entries
Charge collected (PE)
10^0
10^1
10^2
10^3
10^4
10^5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0

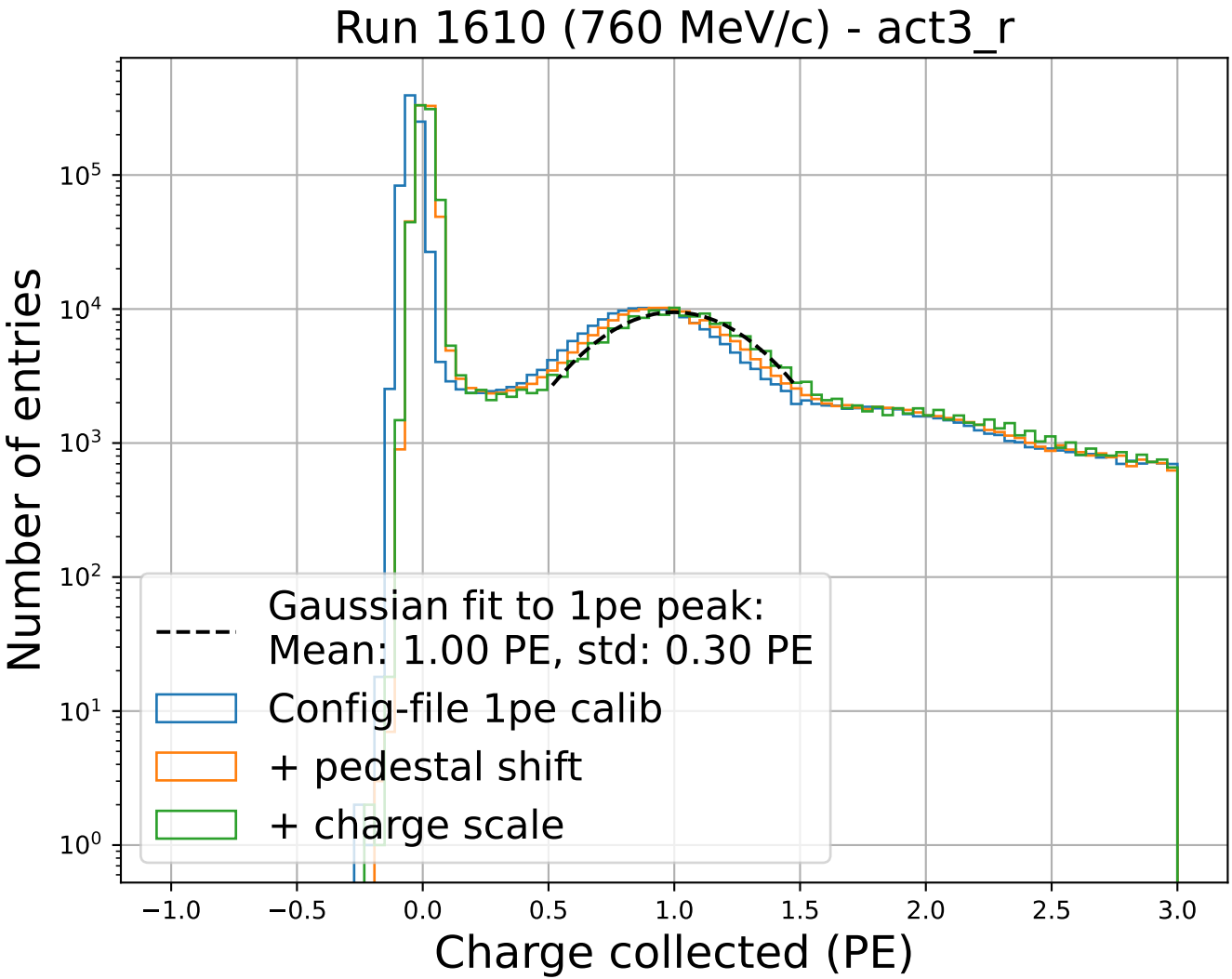

Run 1610 (760 MeV/c) - act1_l
Number of entries
Charge collected (PE)
Gaussian fit to 1pe peak:
Mean: 1.00 PE, std: 0.29 PE
Config-file 1pe calib
+ pedestal shift
+ charge scale

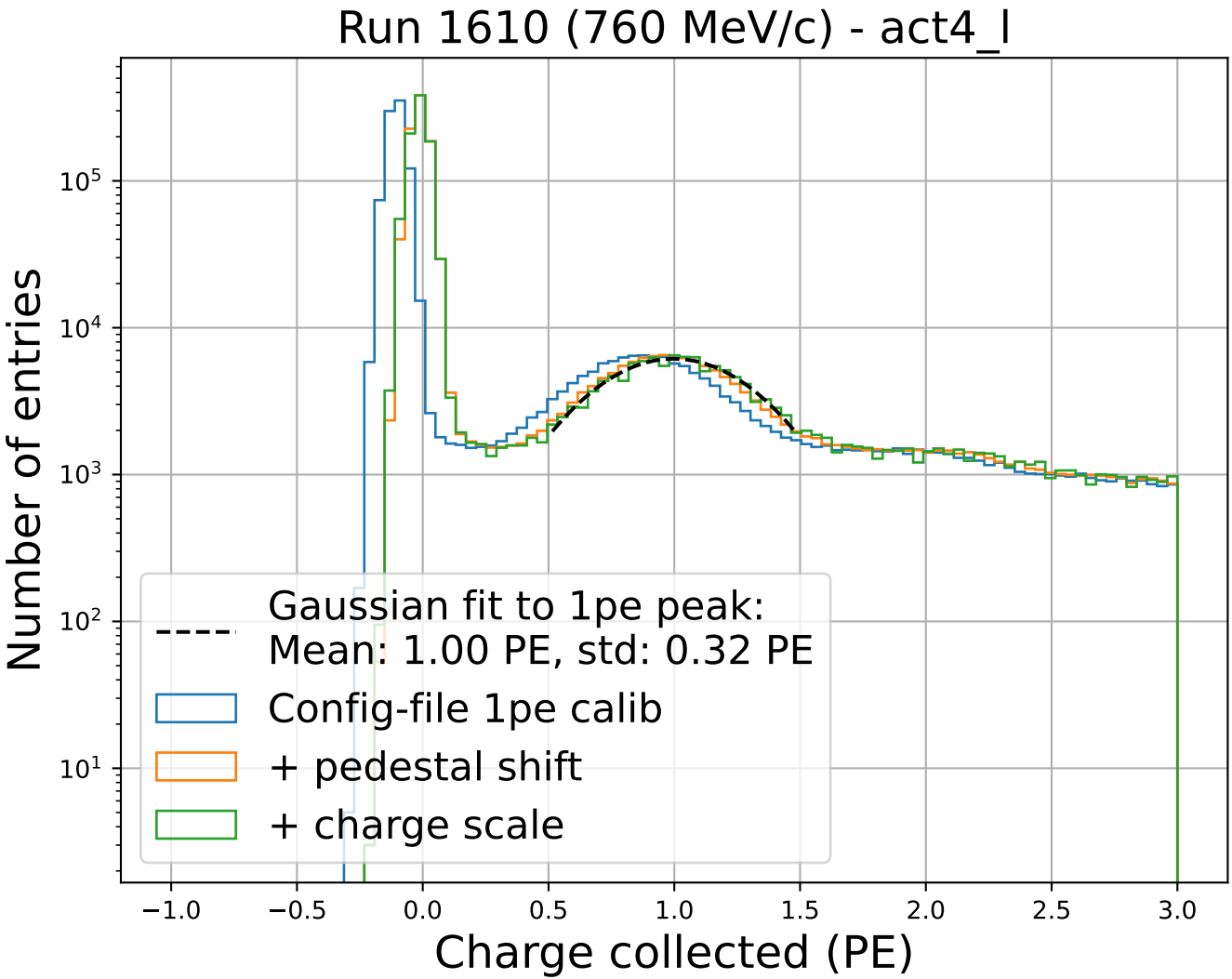

Run 1610 (760 MeV/c) - act1_r
Gaussian fit to 1pe peak:
Mean: 1.00 PE, std: 0.31 PE
Config-file 1pe calib
+ pedestal shift
+ charge scale
Number of entries
Charge collected (PE)
10^6
10^5
10^4
10^3
10^2
10^1
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0

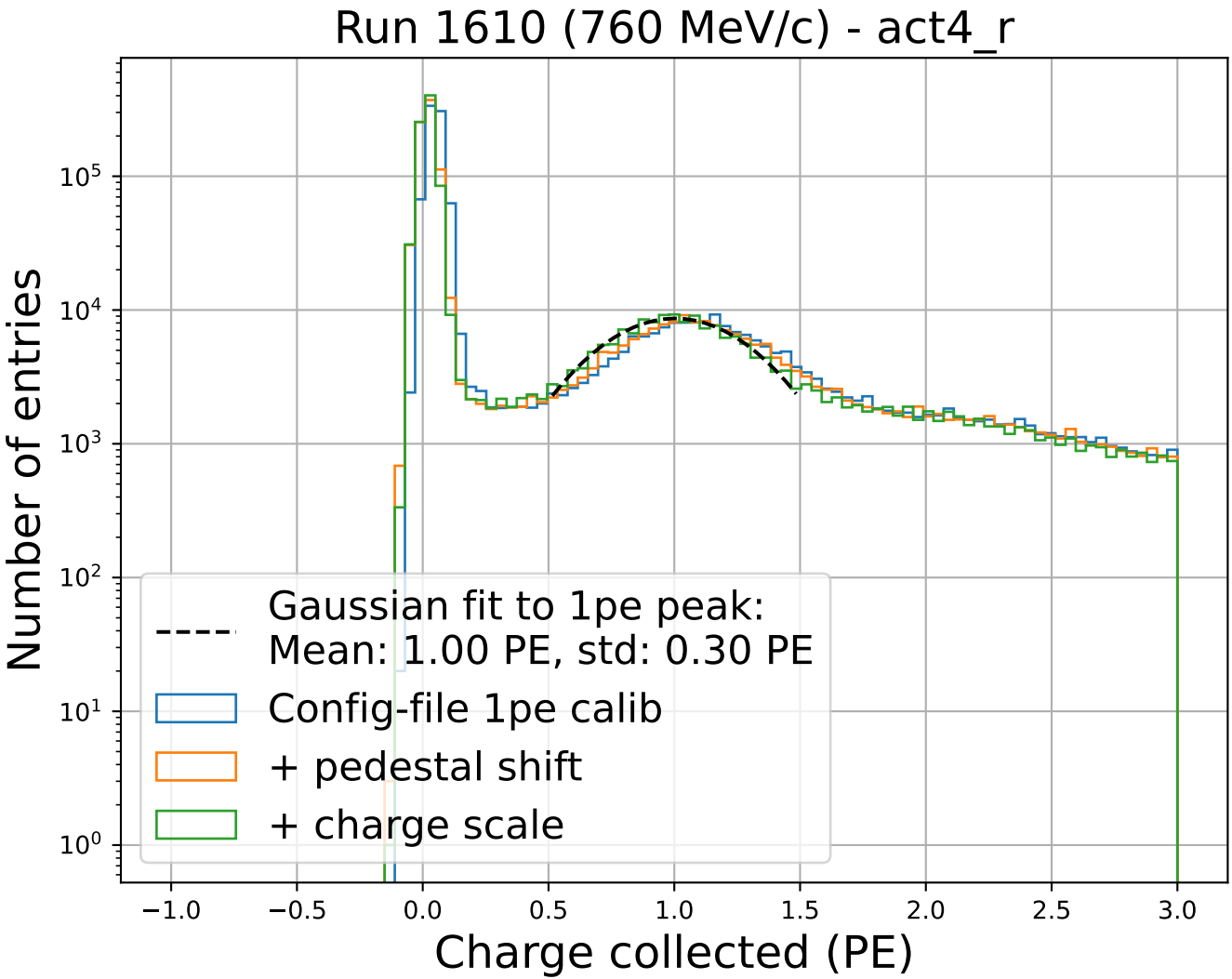

Run 1610 (760 MeV/c) - act2_l
Gaussian fit to 1pe peak:
Mean: 1.00 PE, std: 0.32 PE
Config-file 1pe calib
+ pedestal shift
+ charge scale
Number of entries
Charge collected (PE)
10^5
10^4
10^3
10^2
10^1
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0

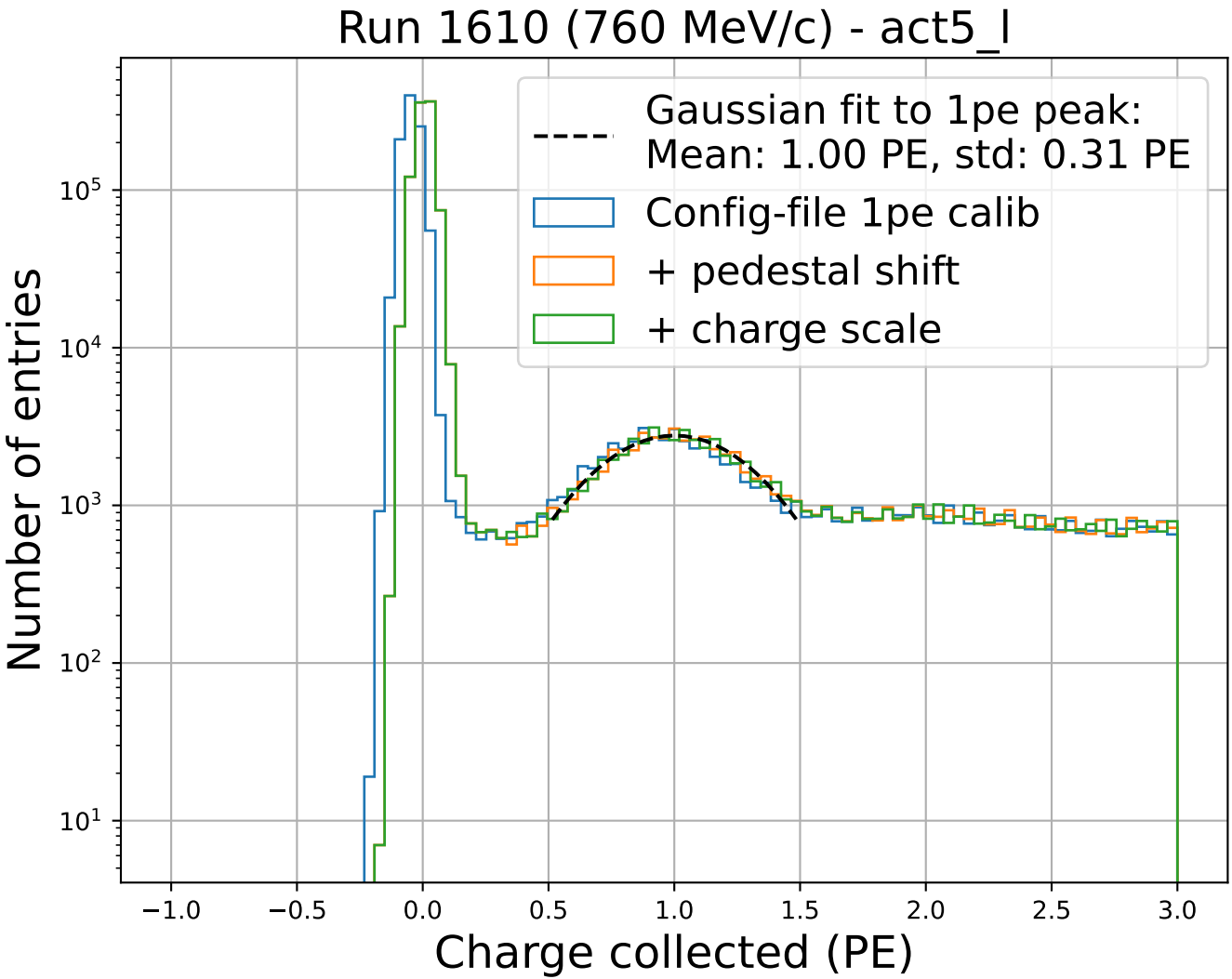

Run 1610 (760 MeV/c) - act2_r
Gaussian fit to 1pe peak:
Mean: 1.00 PE, std: 0.35 PE
Config-file 1pe calib
+ pedestal shift
+ charge scale
Number of entries
Charge collected (PE)
10^0
10^1
10^2
10^3
10^4
10^5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0

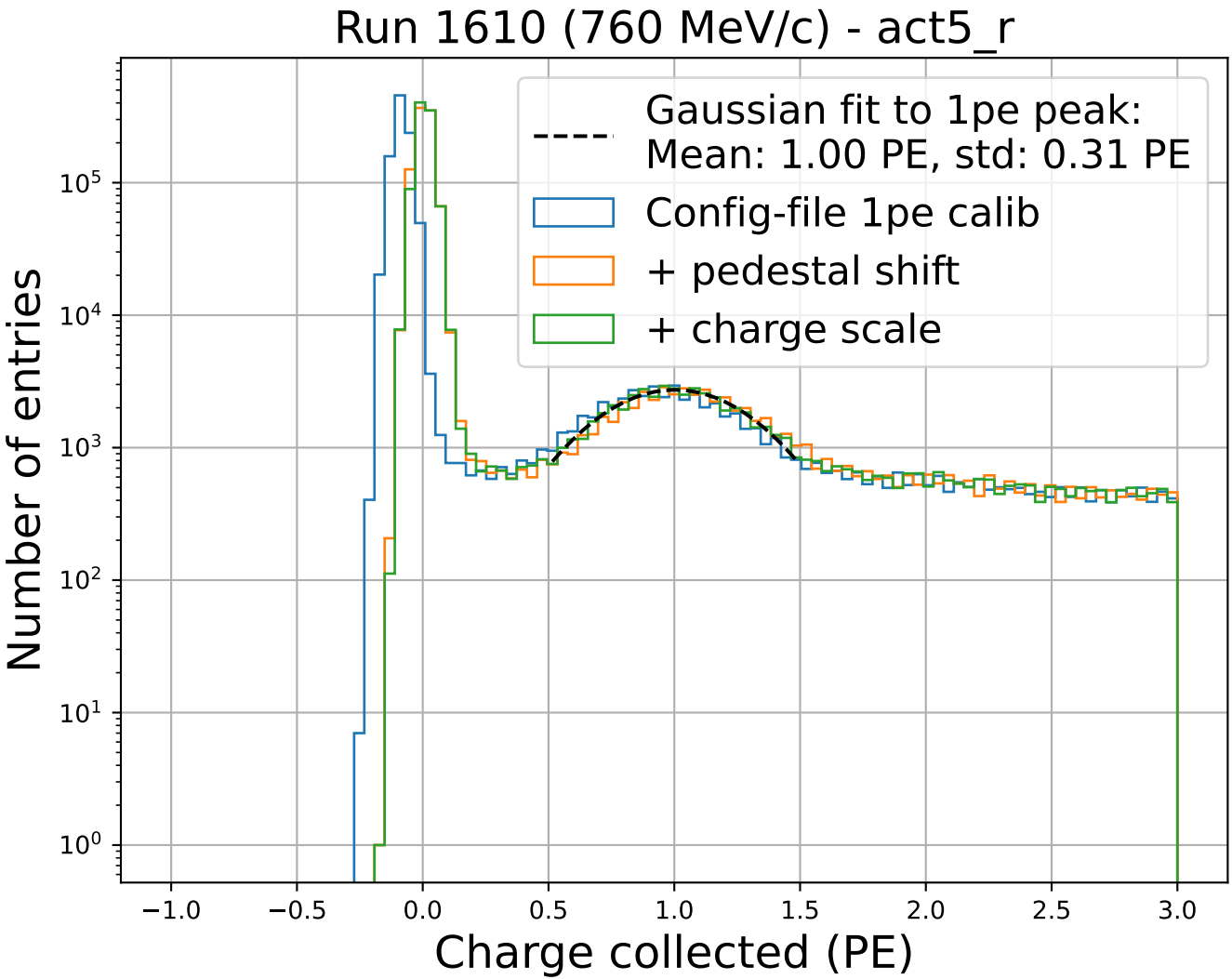

Run 1610 (760 MeV/c) - act3_l
Number of entries
Charge collected (PE)
Gaussian fit to 1pe peak:
Mean: 0.99 PE, std: 0.30 PE
Config-file 1pe calib
+ pedestal shift
+ charge scale

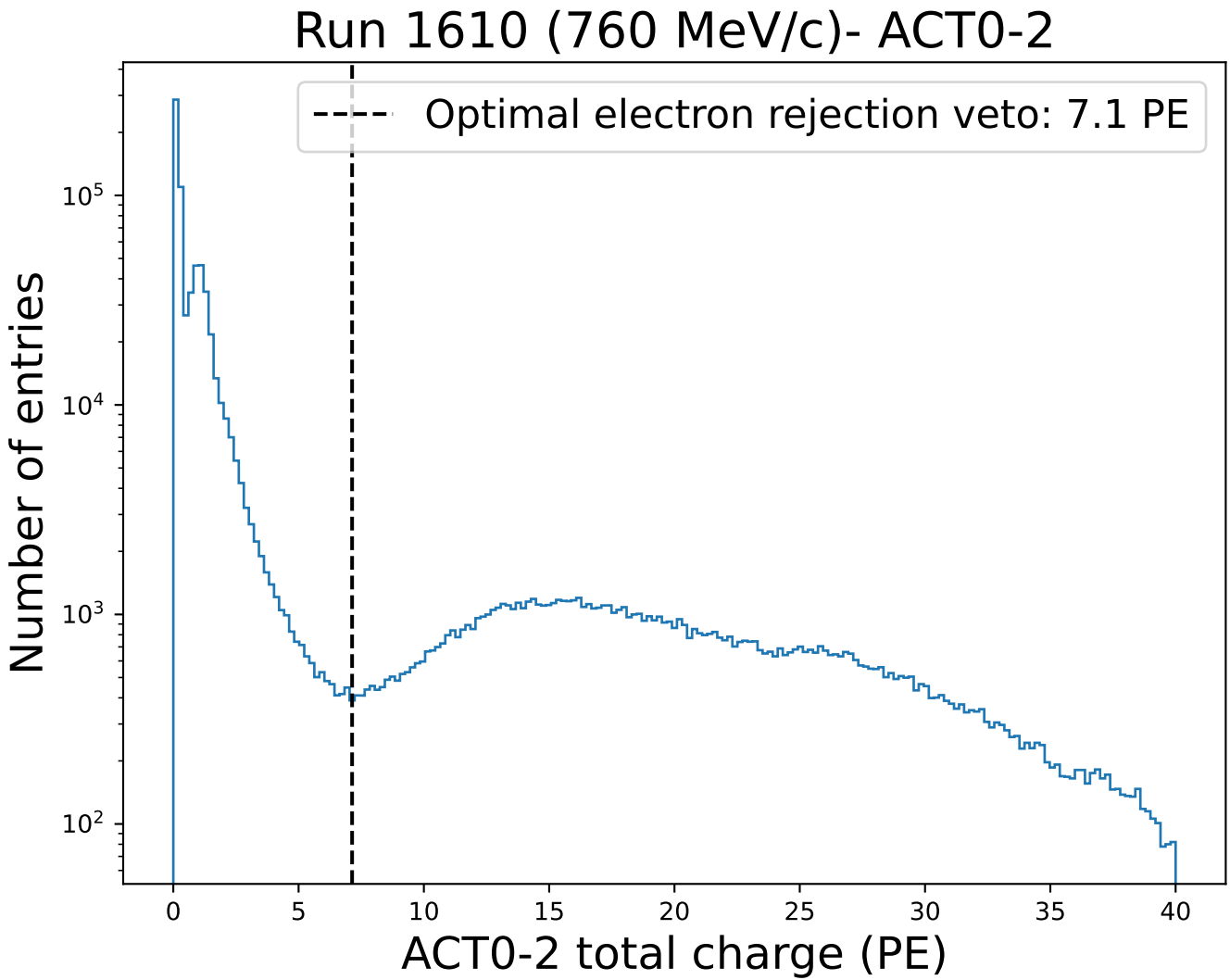

Run 1610 (760 MeV/c) - act3_r
Number of entries
Charge collected (PE)
Gaussian fit to 1pe peak:
Mean: 1.00 PE, std: 0.30 PE
Config-file 1pe calib
+ pedestal shift
+ charge scale
10^0
10^1
10^2
10^3
10^4
10^5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0

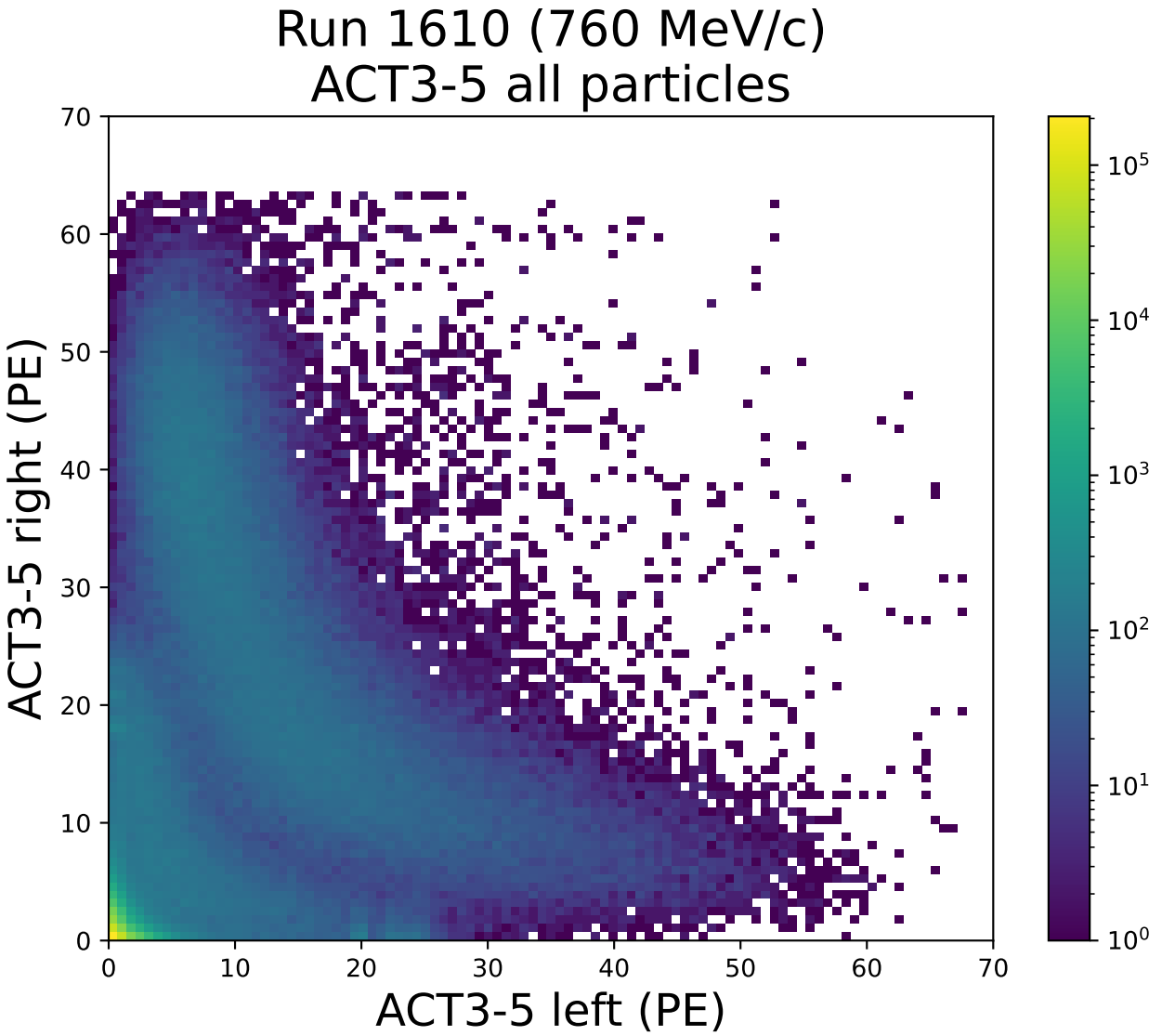

Run 1610 (760 MeV/c) - act4_l
Number of entries
Charge collected (PE)
Gaussian fit to 1pe peak:
Mean: 1.00 PE, std: 0.32 PE
Config-file 1pe calib
+ pedestal shift
+ charge scale
10^5
10^4
10^3
10^2
10^1
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0

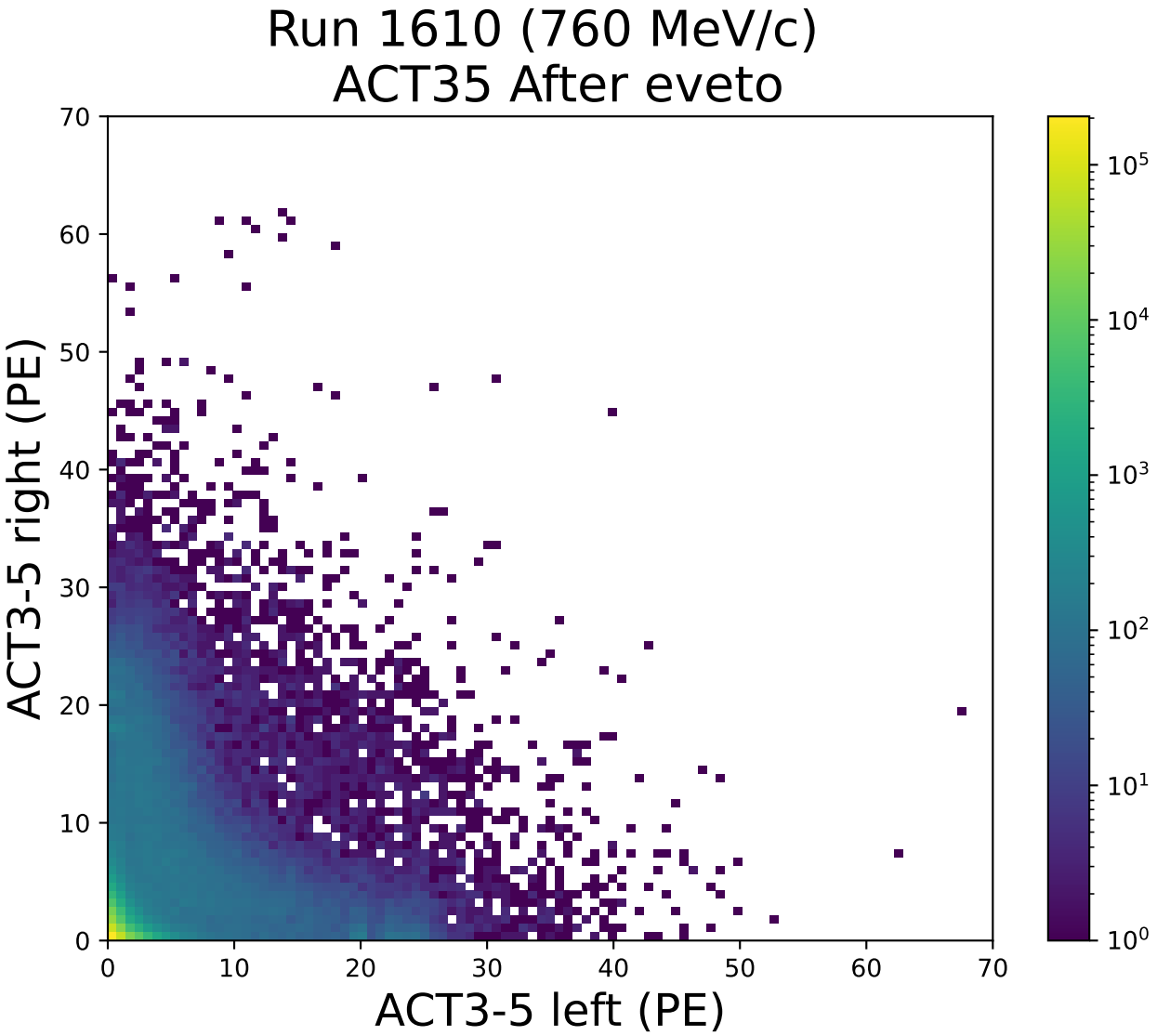

Run 1610 (760 MeV/c) - act4_r
Number of entries
Charge collected (PE)
Gaussian fit to 1pe peak:
Mean: 1.00 PE, std: 0.30 PE
Config-file 1pe calib
+ pedestal shift
+ charge scale

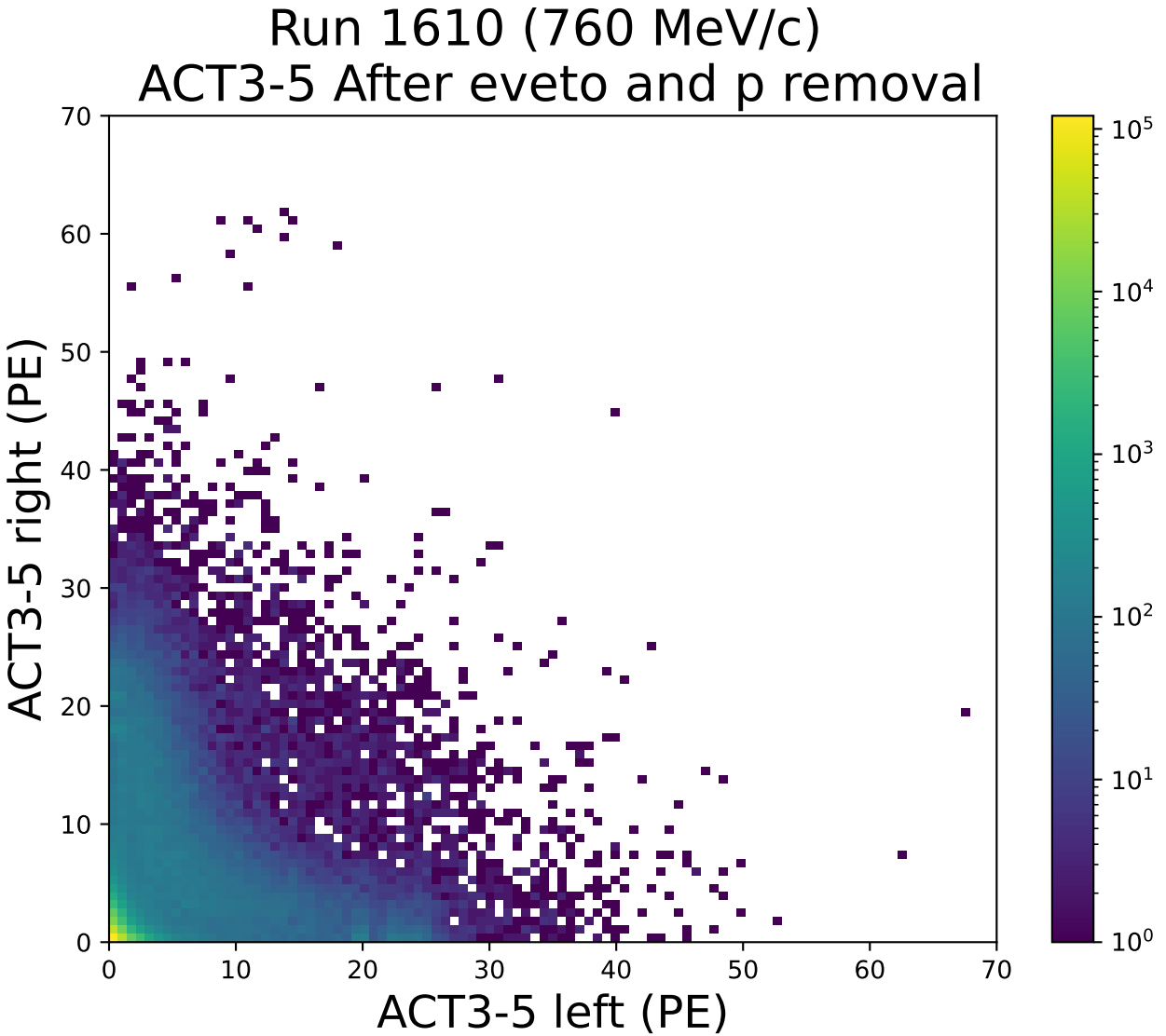

Run 1610 (760 MeV/c) - act5_l
Gaussian fit to 1pe peak:
Mean: 1.00 PE, std: 0.31 PE
Config-file 1pe calib
+ pedestal shift
+ charge scale
Number of entries
Charge collected (PE)
10^5
10^4
10^3
10^2
10^1
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0

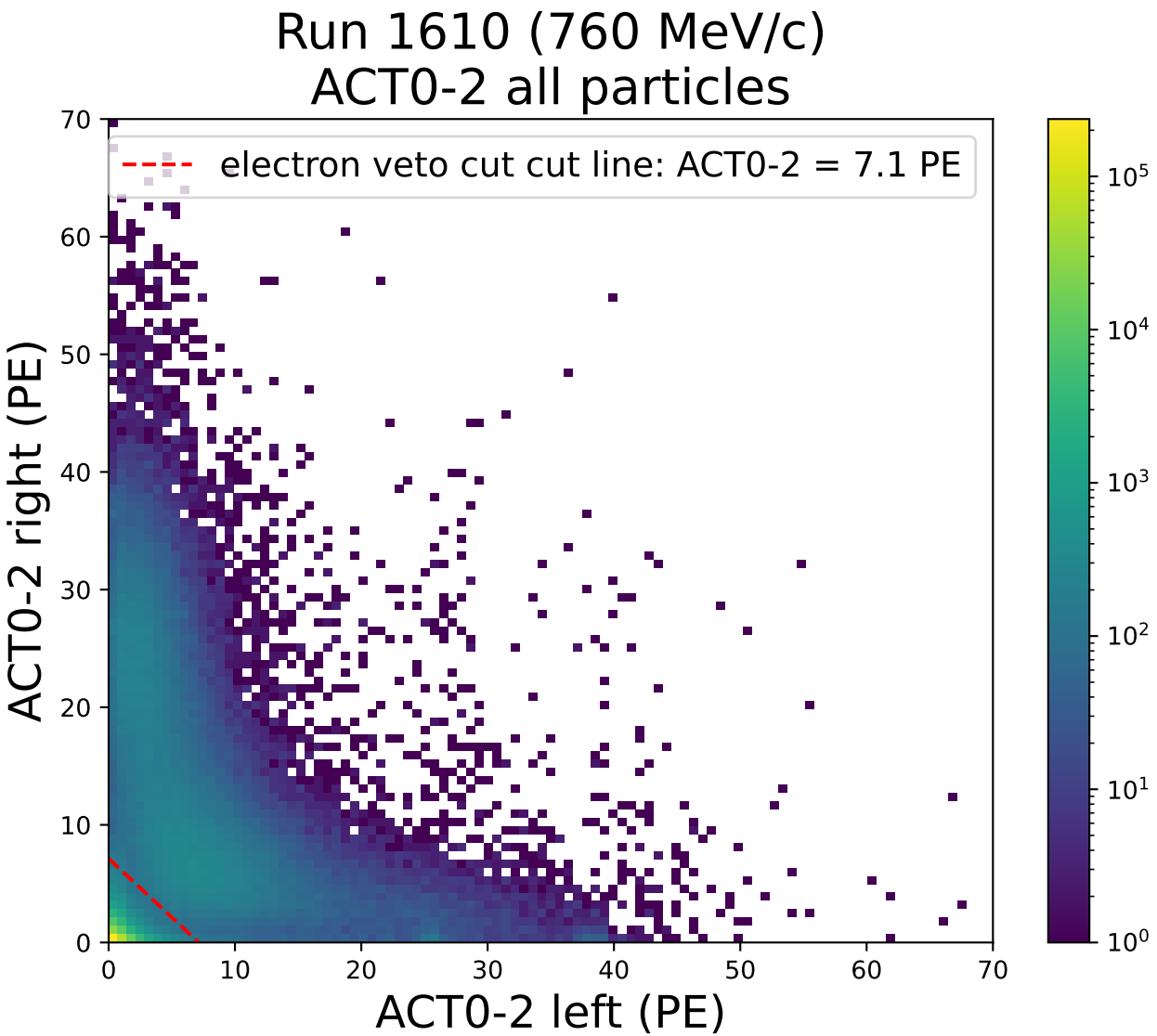

Run 1610 (760 MeV/c) - act5_r
Gaussian fit to 1pe peak:
Mean: 1.00 PE, std: 0.31 PE
Config-file 1pe calib
+ pedestal shift
+ charge scale
Number of entries
Charge collected (PE)
10^0
10^1
10^2
10^3
10^4
10^5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0

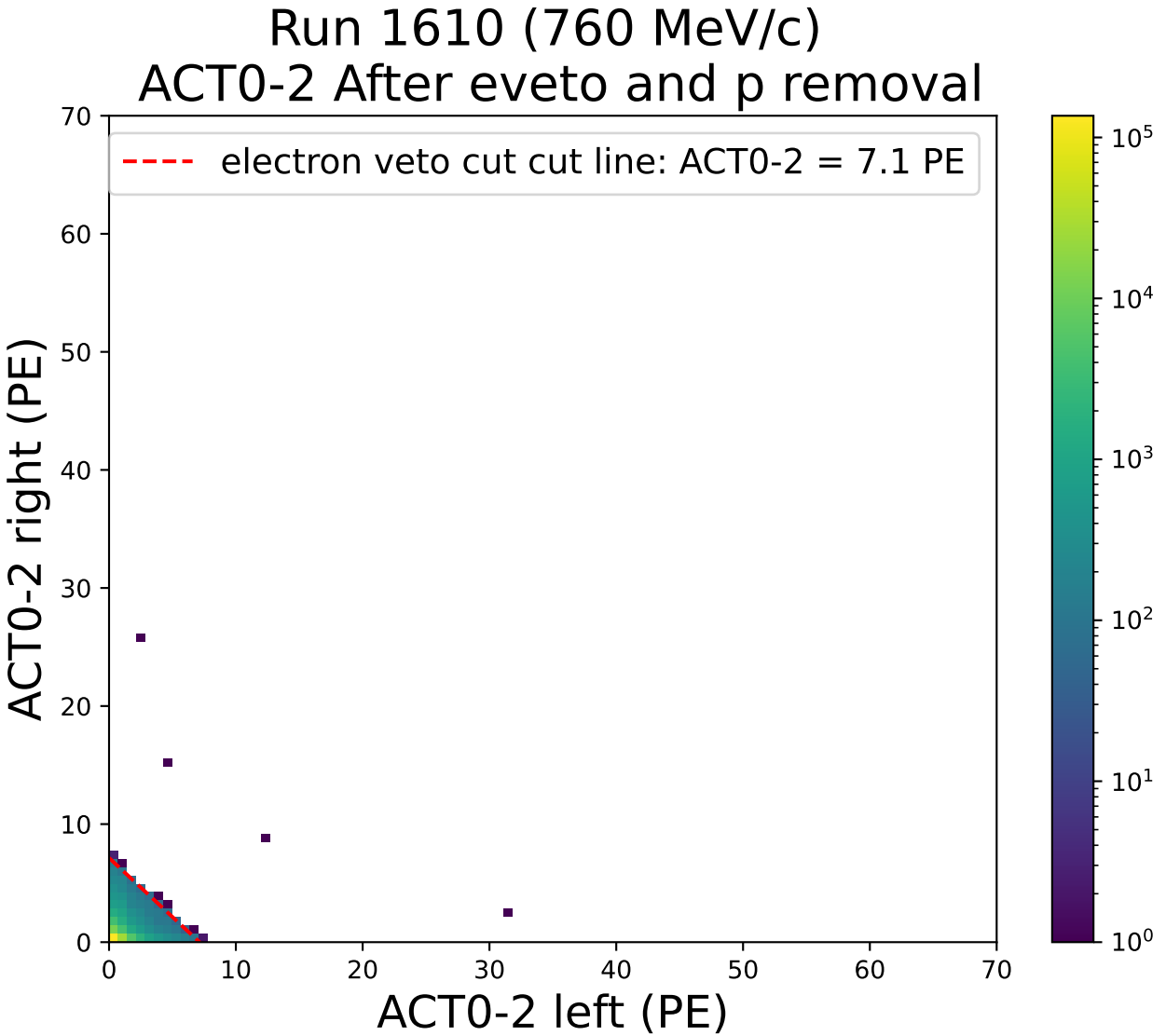

Run 1610 (760 MeV/c)- ACT0-2
Optimal electron rejection veto: 7.1 PE
Number of entries
10^5
10^4
10^3
10^2
0
5
10
15
20
25
30
35
40
ACT0-2 total charge (PE)

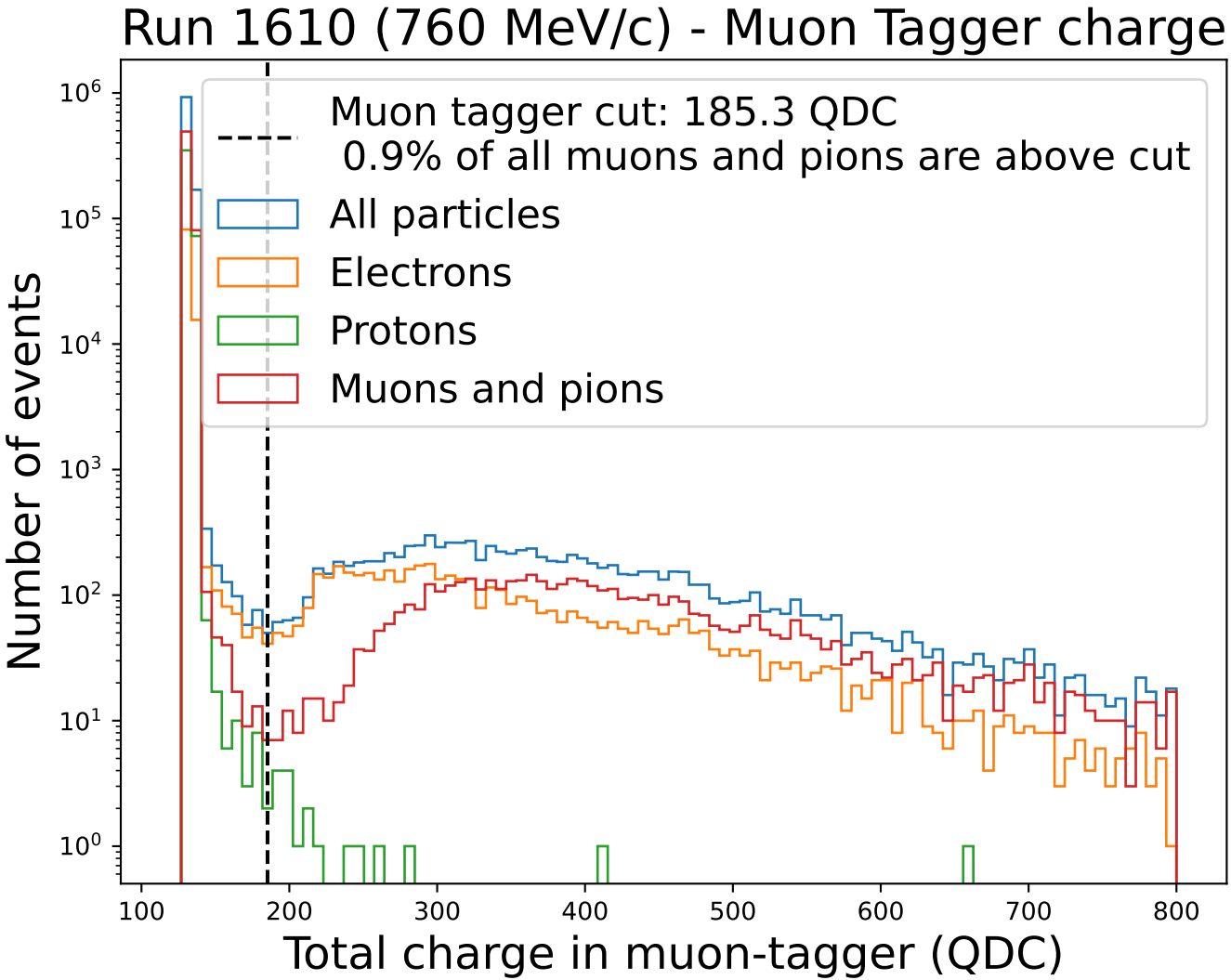

Run 1610 (760 MeV/c)
ACT3-5 all particles
ACT3-5 right (PE)
ACT3-5 left (PE)
0
10
20
30
40
50
60
70
10^0
10^1
10^2
10^3
10^4
10^5

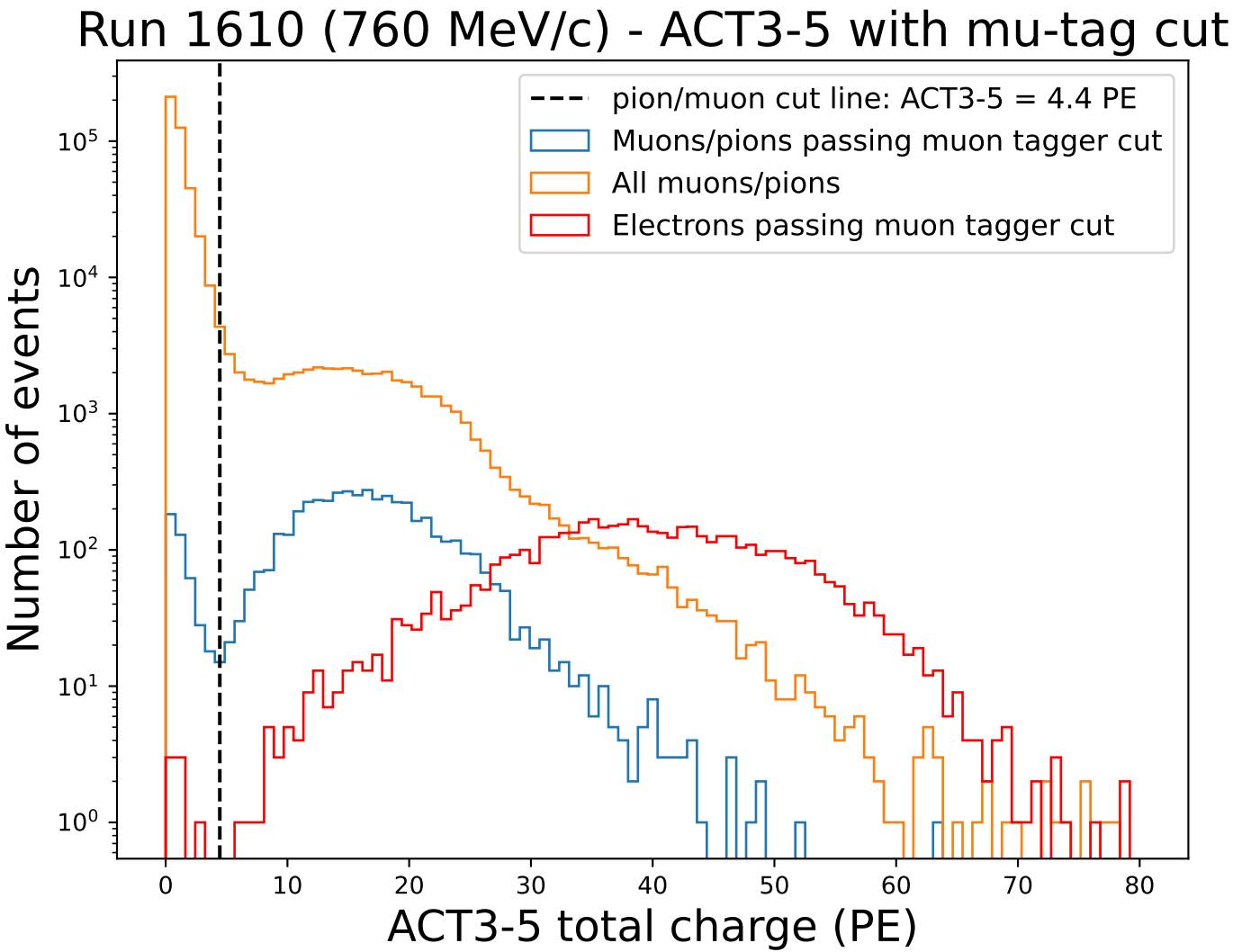

Run 1610 (760 MeV/c)
ACT35 After eveto
ACT3-5 right (PE)
ACT3-5 left (PE)
0
10
20
30
40
50
60
70
10^0
10^1
10^2
10^3
10^4
10^5

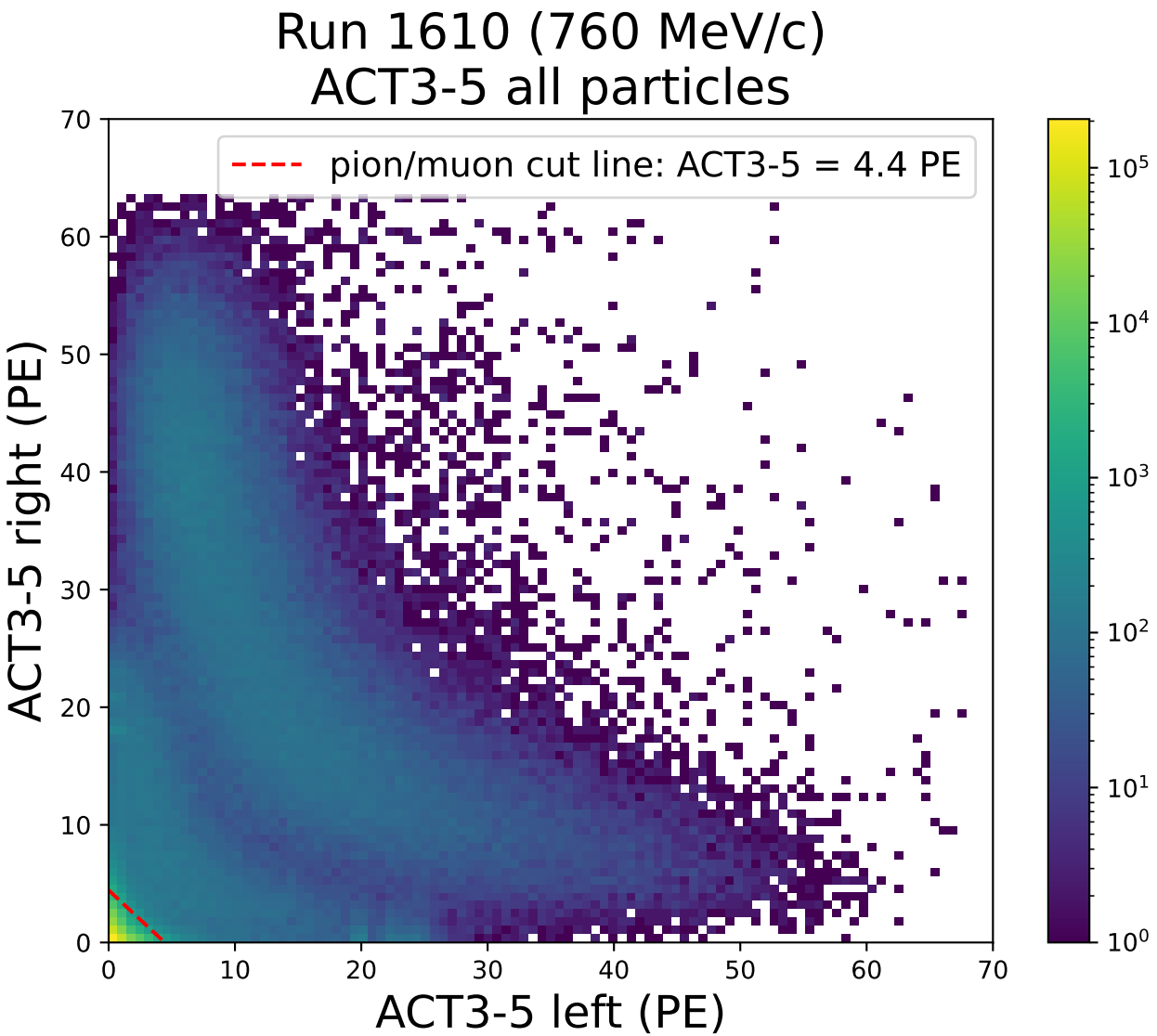

Run 1610 (760 MeV/c)
ACT3-5 After eveto and p removal
ACT3-5 right (PE)
ACT3-5 left (PE)
0
10
20
30
40
50
60
70
10^0
10^1
10^2
10^3
10^4
10^5

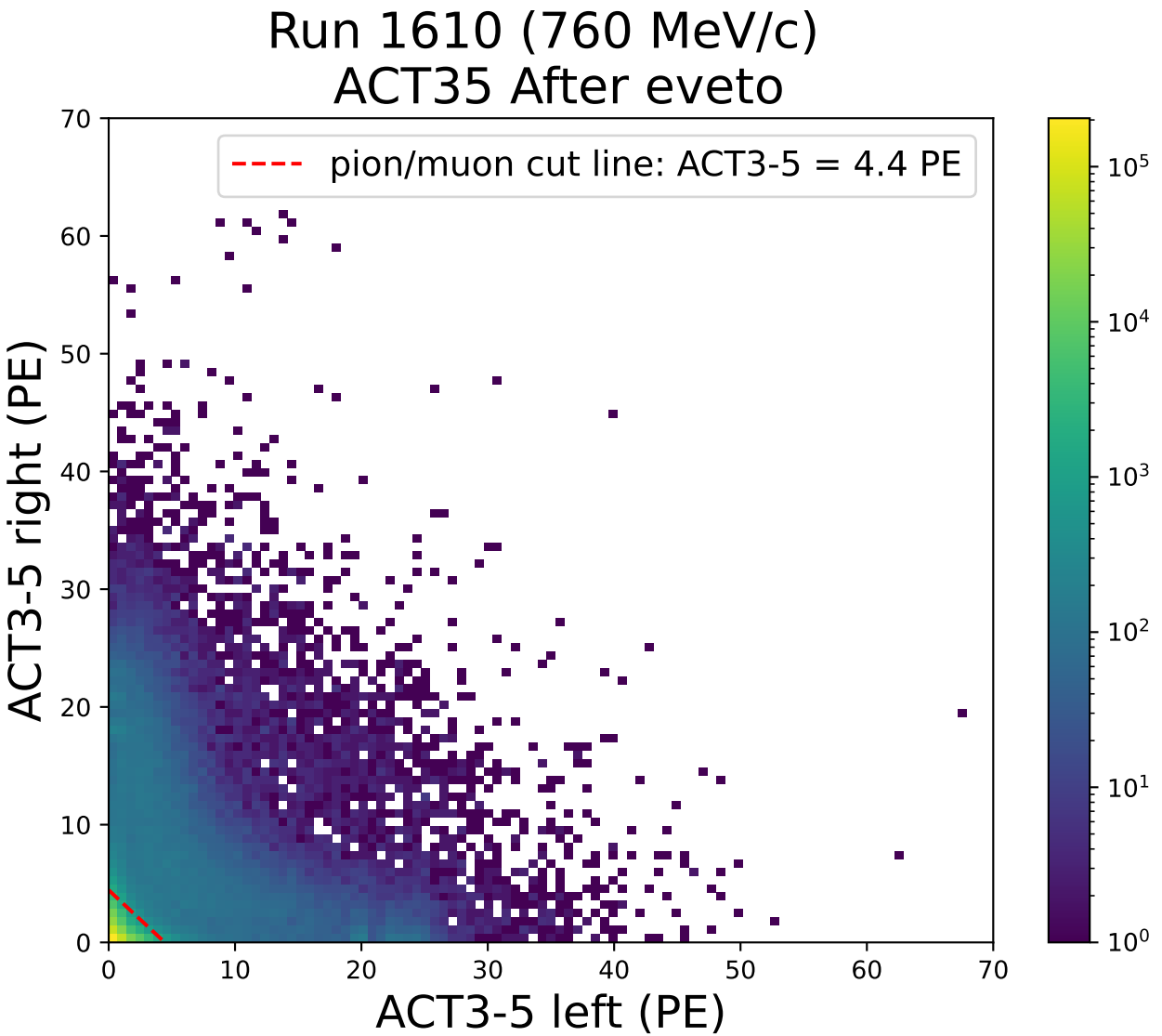

Run 1610 (760 MeV/c)
ACT0-2 all particles
electron veto cut cut line: ACT0-2 = 7.1 PE
ACT0-2 right (PE)
ACT0-2 left (PE)
0
10
20
30
40
50
60
70
10^0
10^1
10^2
10^3
10^4
10^5

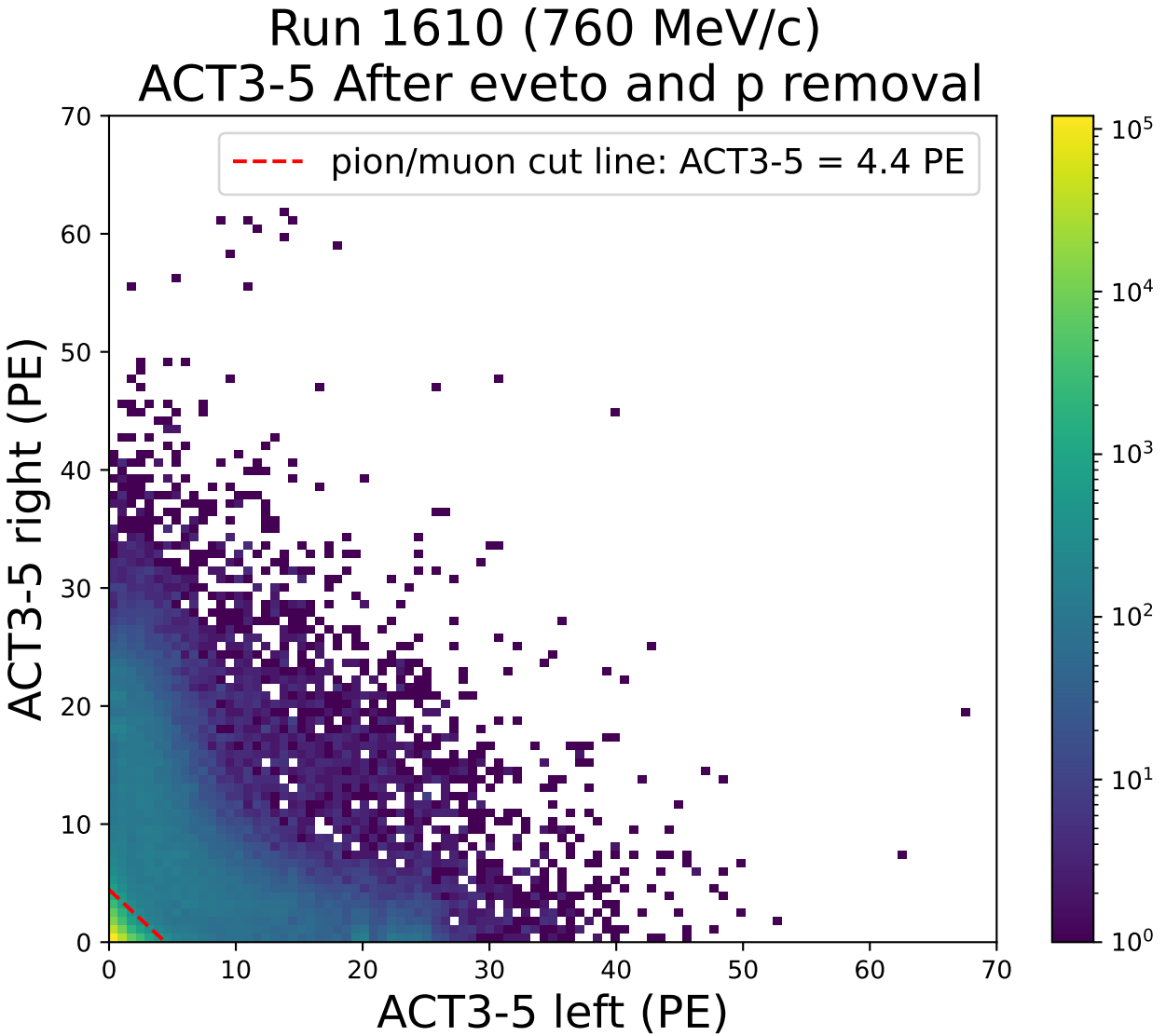

Run 1610 (760 MeV/c)
ACT0-2 After eveto and p removal
electron veto cut cut line: ACT0-2 = 7.1 PE
ACT0-2 right (PE)
ACT0-2 left (PE)
0
10
20
30
40
50
60
70
10^0
10^1
10^2
10^3
10^4
10^5

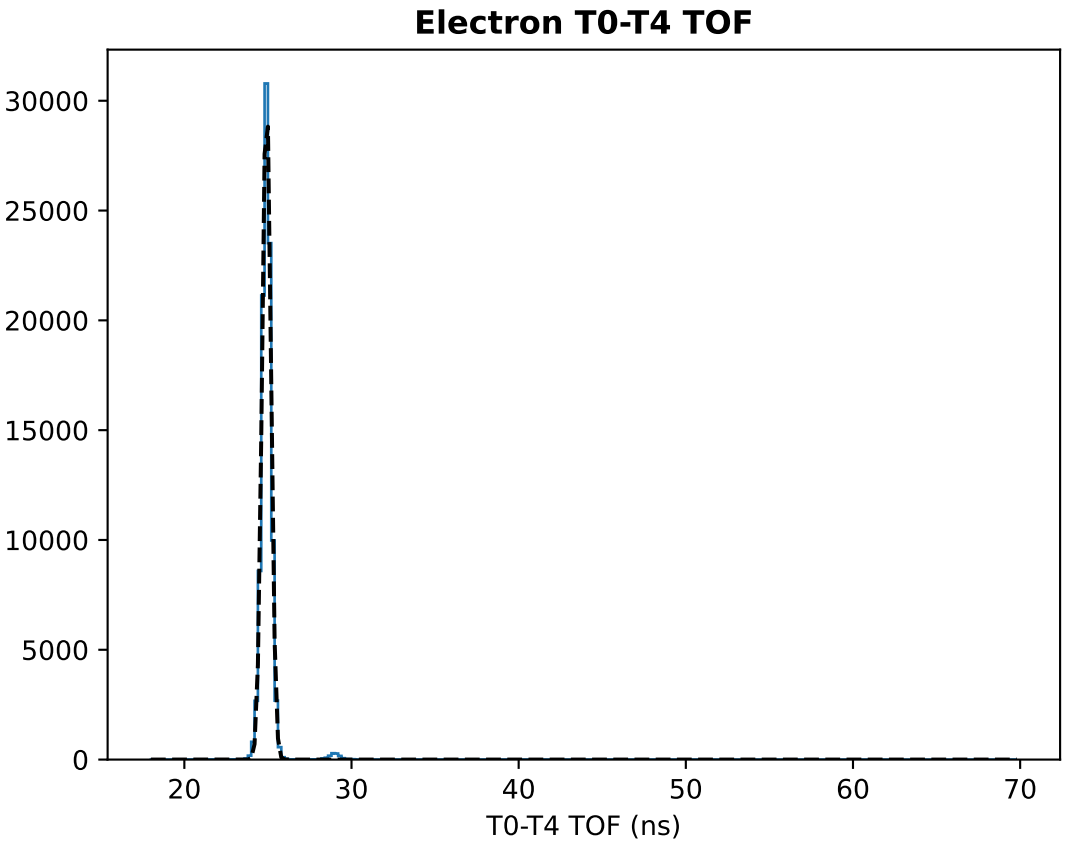

Run 1610 (760 MeV/c) - Muon Tagger charge
Muon tagger cut: 185.3 QDC
0.9% of all muons and pions are above cut
All particles
Electrons
Protons
Muons and pions
Number of events
Total charge in muon-tagger (QDC)
10^0
10^1
10^2
10^3
10^4
10^5
10^6
100
200
300
400
500
600
700
800

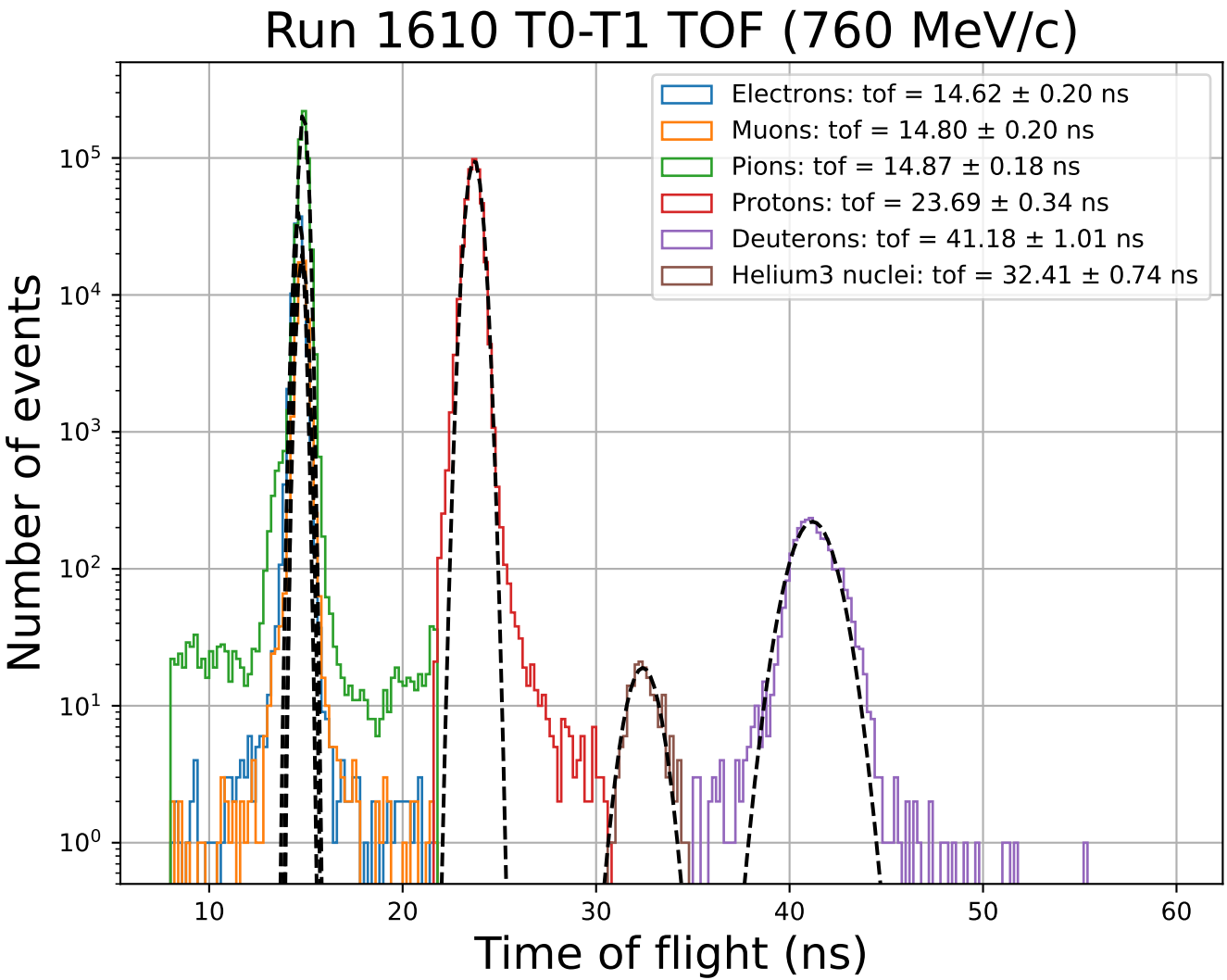

Run 1610 (760 MeV/c) - ACT3-5 with mu-tag cut
pion/muon cut line: ACT3-5 = 4.4 PE
Muons/pions passing muon tagger cut
All muons/pions
Electrons passing muon tagger cut
Number of events
ACT3-5 total charge (PE)
10^0
10^1
10^2
10^3
10^4
10^5
0
10
20
30
40
50
60
70
80

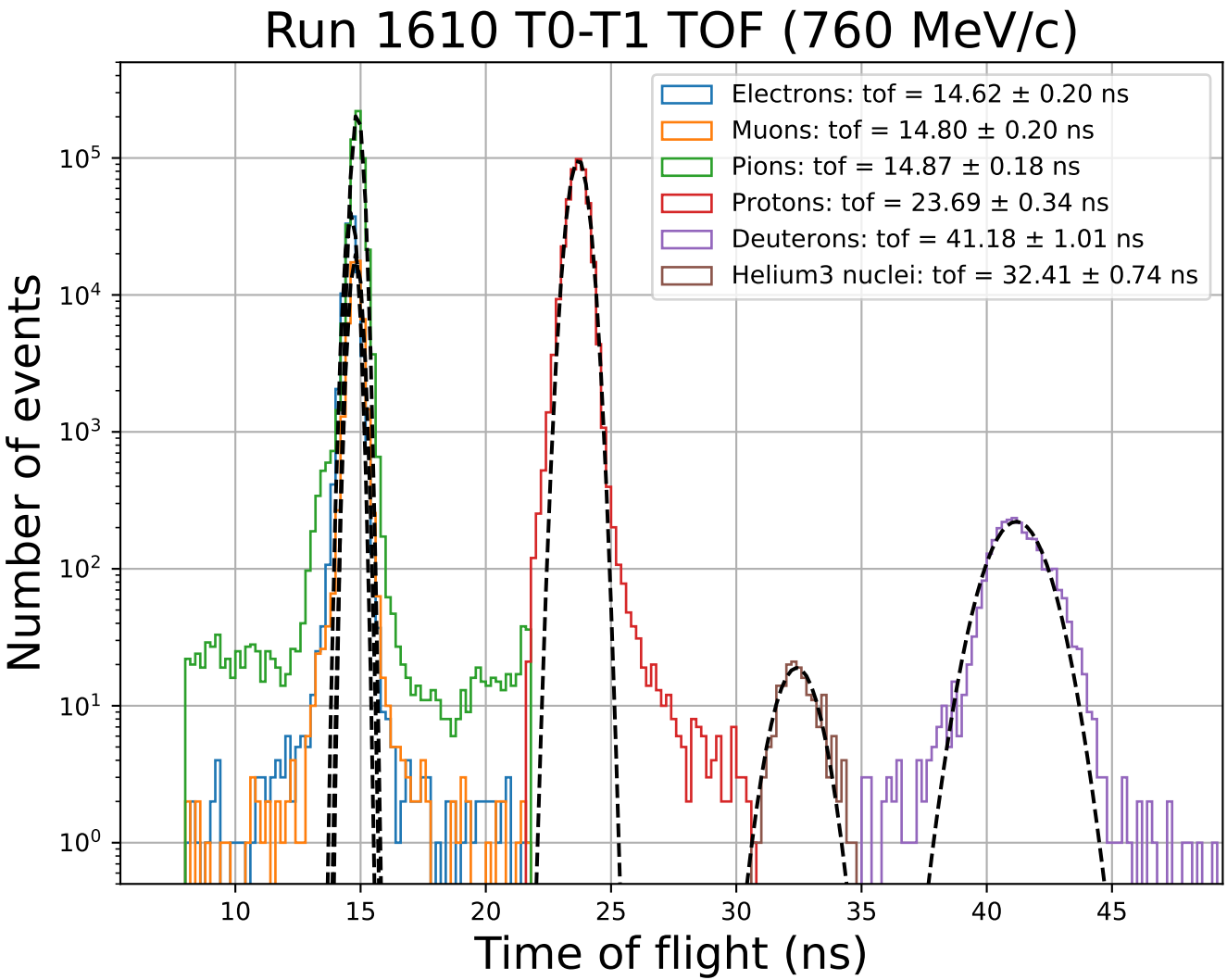

Run 1610 (760 MeV/c)
ACT3-5 all particles
pion/muon cut line: ACT3-5 = 4.4 PE
ACT3-5 right (PE)
ACT3-5 left (PE)
0
10
20
30
40
50
60
70
10^0
10^1
10^2
10^3
10^4
10^5

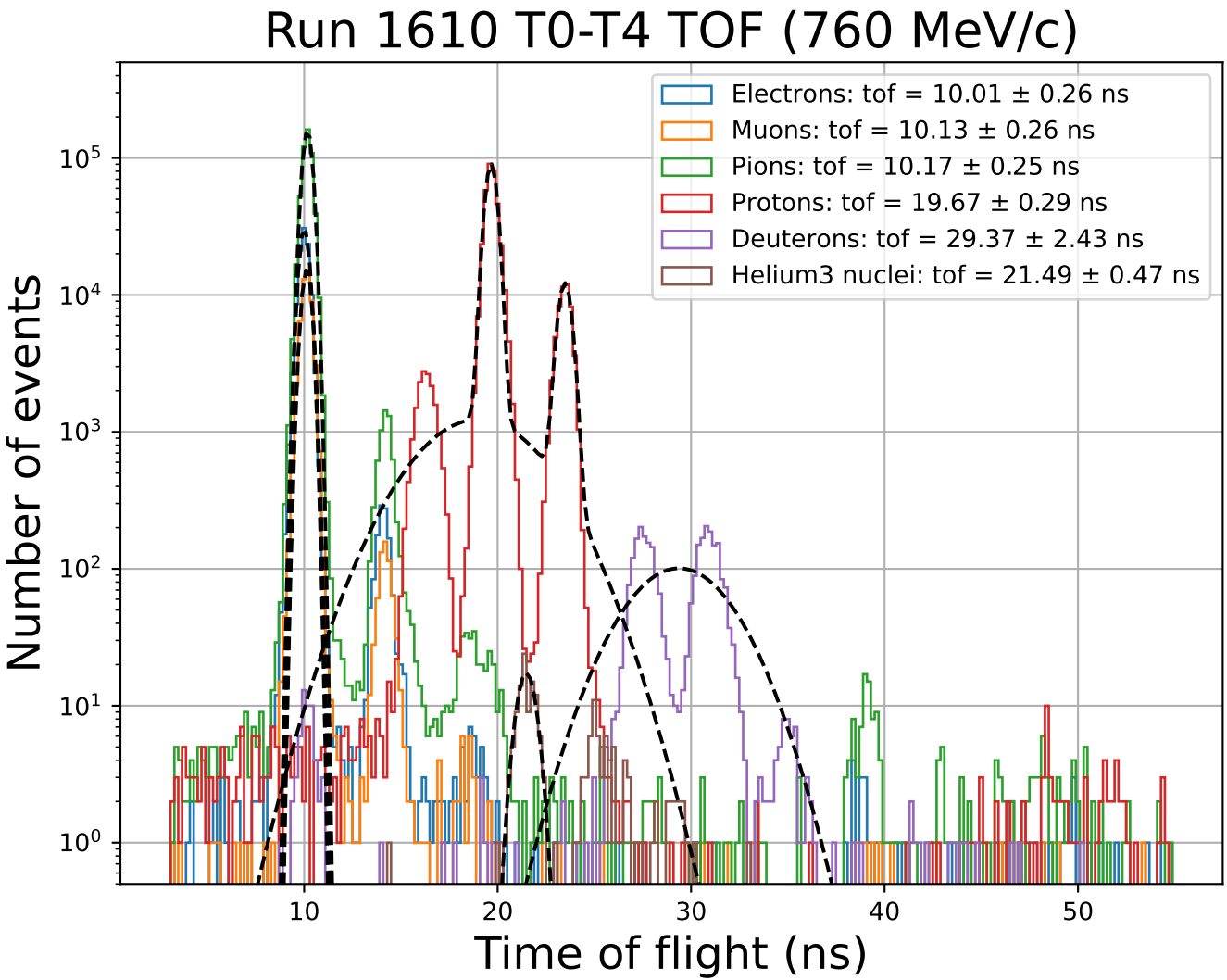

Run 1610 (760 MeV/c)
ACT35 After eveto
pion/muon cut line: ACT3-5 = 4.4 PE
ACT3-5 right (PE)
ACT3-5 left (PE)
0
10
20
30
40
50
60
70
10^0
10^1
10^2
10^3
10^4
10^5

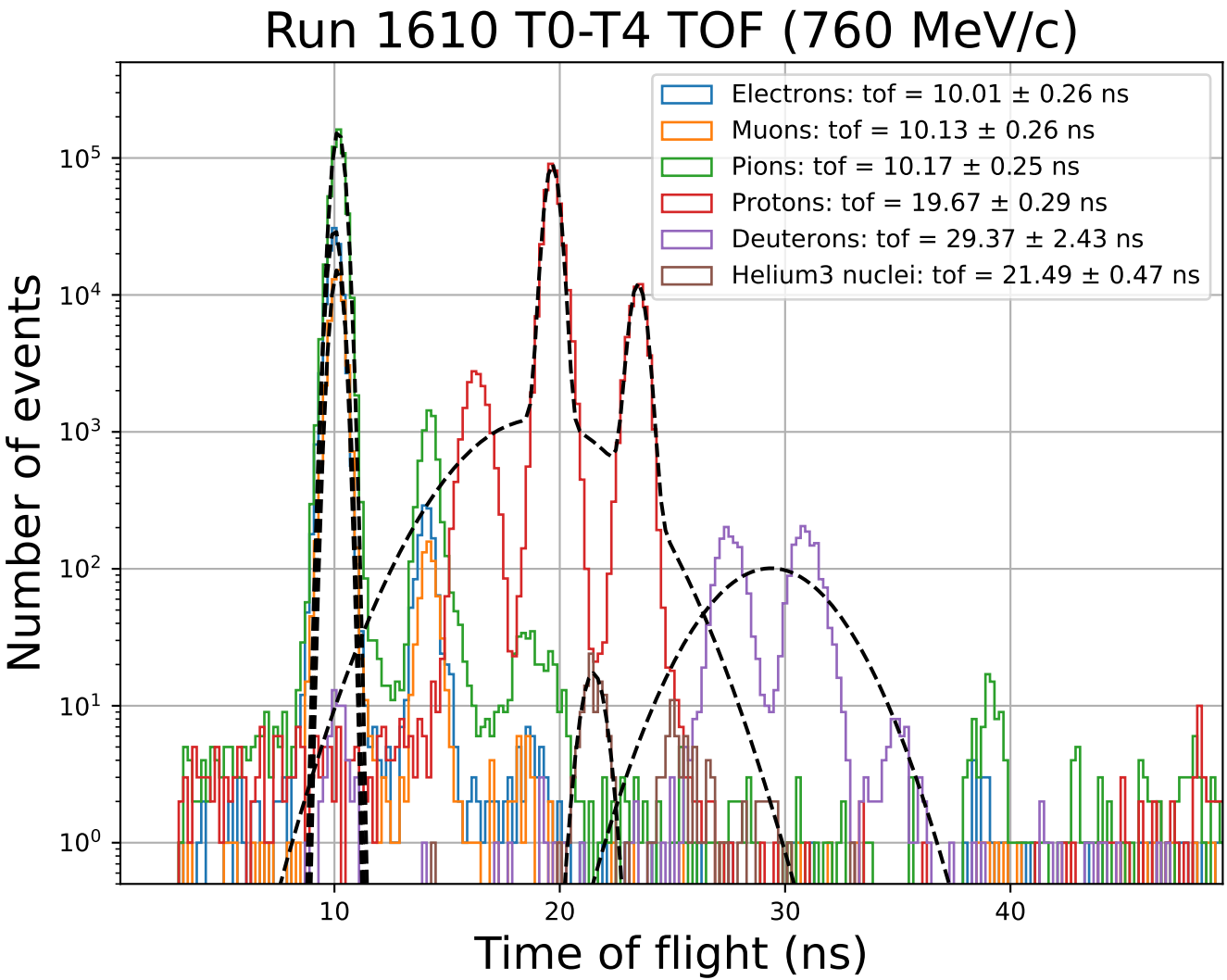

Run 1610 (760 MeV/c)
ACT3-5 After eveto and p removal
pion/muon cut line: ACT3-5 = 4.4 PE
ACT3-5 right (PE)
ACT3-5 left (PE)
0
10
20
30
40
50
60
70
10^0
10^1
10^2
10^3
10^4
10^5

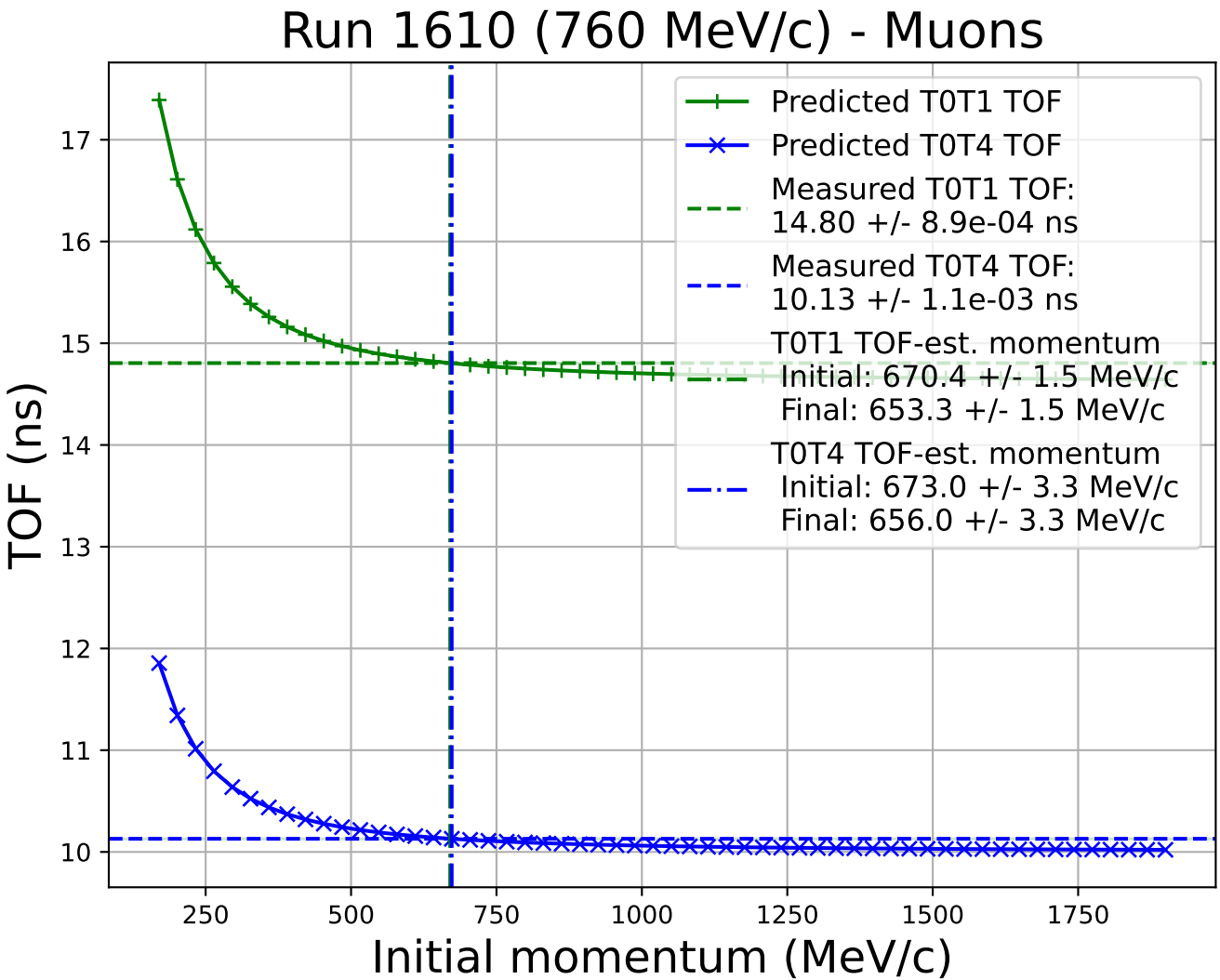

Electron T0-T4 TOF
30000
25000
20000
15000
10000
5000
0
20
30
40
50
60
70
T0-T4 TOF (ns)

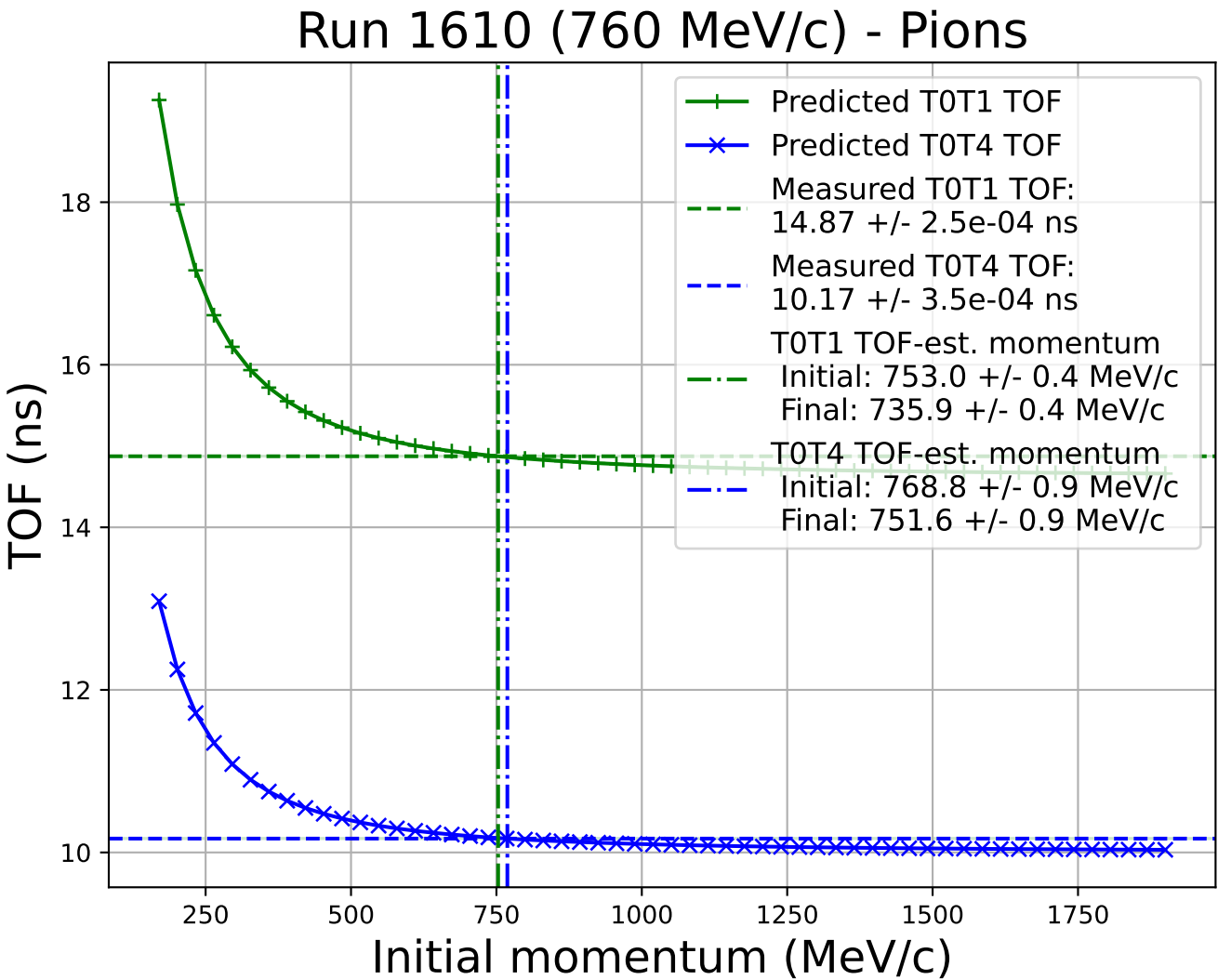

Run 1610 T0-T1 TOF (760 MeV/c)
Electrons: tof = 14.62 ± 0.20 ns
Muons: tof = 14.80 ± 0.20 ns
Pions: tof = 14.87 ± 0.18 ns
Protons: tof = 23.69 ± 0.34 ns
Deuterons: tof = 41.18 ± 1.01 ns
Helium3 nuclei: tof = 32.41 ± 0.74 ns
Number of events
Time of flight (ns)
10
20
30
40
50
60

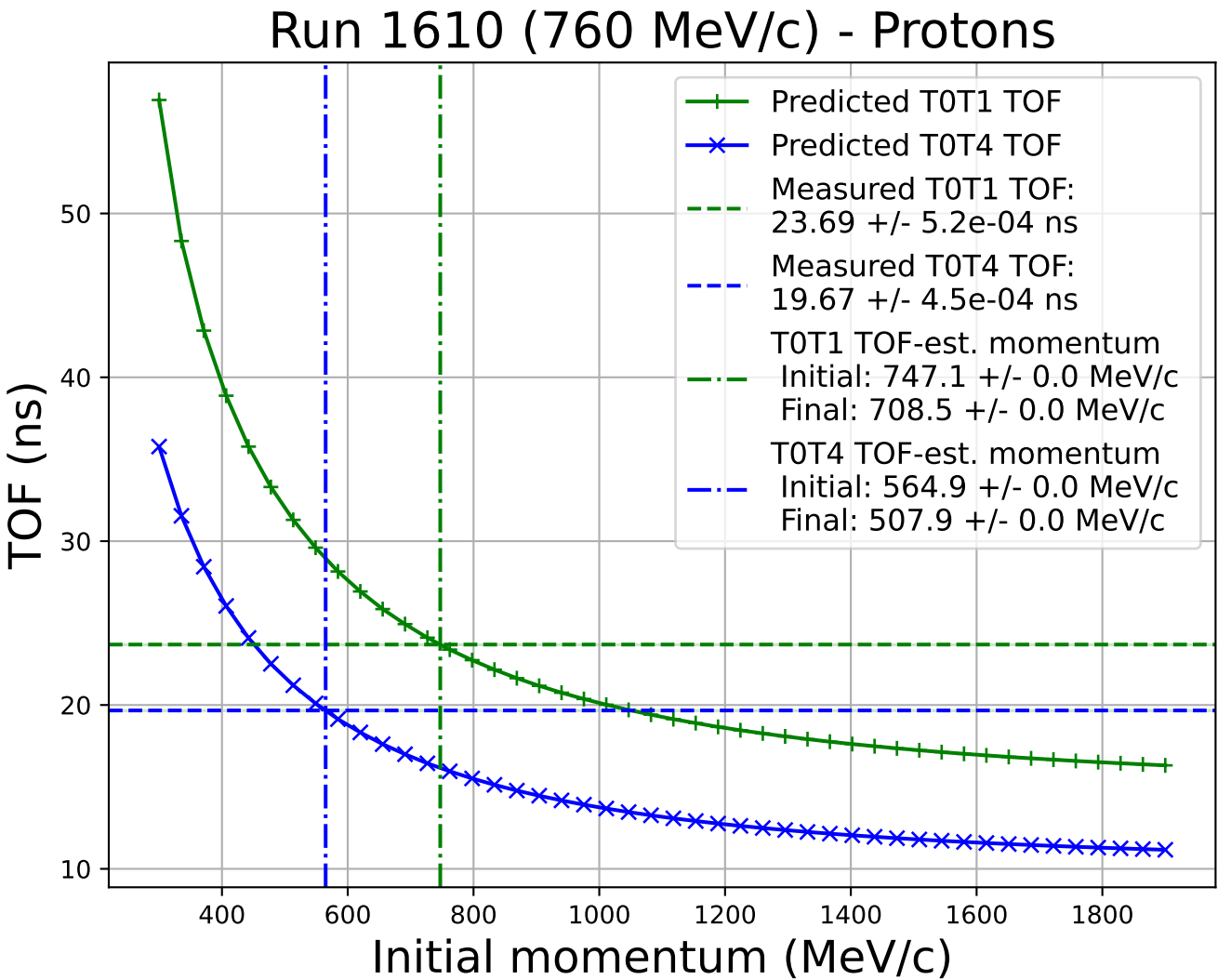

Run 1610 T0-T1 TOF (760 MeV/c)
Electrons: tof = 14.62 ± 0.20 ns
Muons: tof = 14.80 ± 0.20 ns
Pions: tof = 14.87 ± 0.18 ns
Protons: tof = 23.69 ± 0.34 ns
Deuterons: tof = 41.18 ± 1.01 ns
Helium3 nuclei: tof = 32.41 ± 0.74 ns
Number of events
Time of flight (ns)
10^0
10^1
10^2
10^3
10^4
10^5
10
15
20
25
30
35
40
45

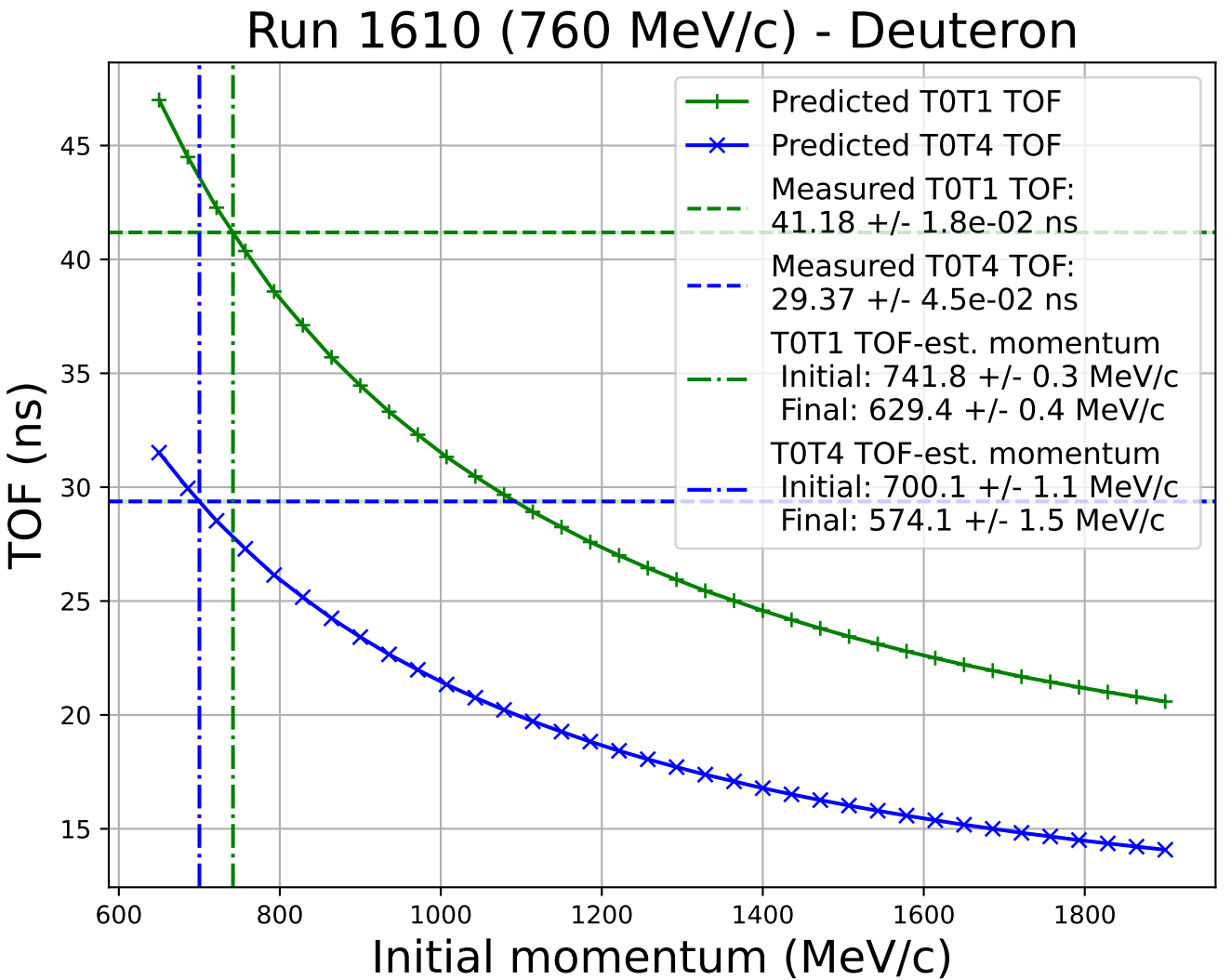

Run 1610 T0-T4 TOF (760 MeV/c)
Electrons: tof = 10.01 ± 0.26 ns
Muons: tof = 10.13 ± 0.26 ns
Pions: tof = 10.17 ± 0.25 ns
Protons: tof = 19.67 ± 0.29 ns
Deuterons: tof = 29.37 ± 2.43 ns
Helium3 nuclei: tof = 21.49 ± 0.47 ns
Number of events
Time of flight (ns)
10^0
10^1
10^2
10^3
10^4
10^5
10
20
30
40
50

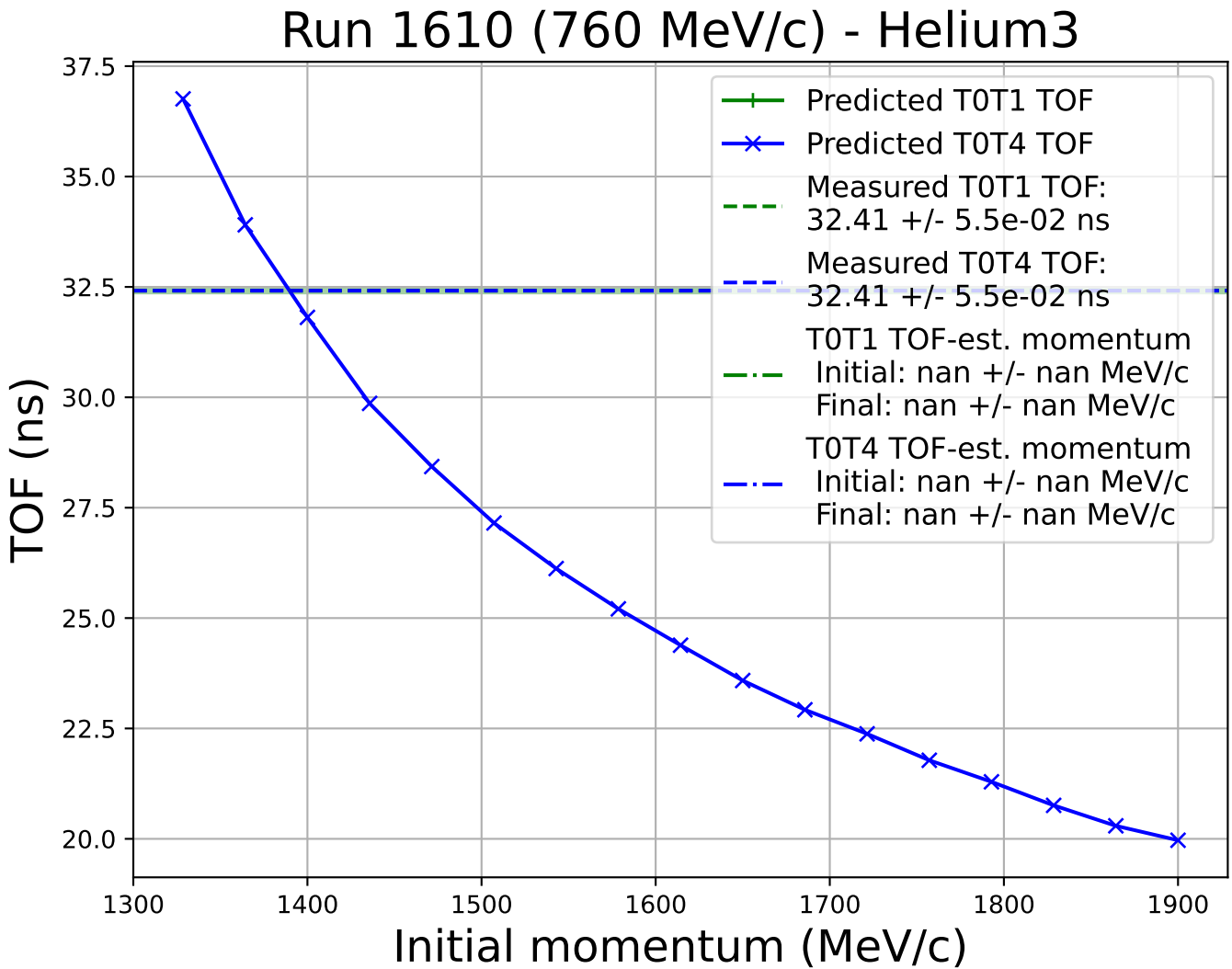

Run 1610 T0-T4 TOF (760 MeV/c)
Electrons: tof = 10.01 ± 0.26 ns
Muons: tof = 10.13 ± 0.26 ns
Pions: tof = 10.17 ± 0.25 ns
Protons: tof = 19.67 ± 0.29 ns
Deuterons: tof = 29.37 ± 2.43 ns
Helium3 nuclei: tof = 21.49 ± 0.47 ns
Number of events
Time of flight (ns)
10
20
30
40

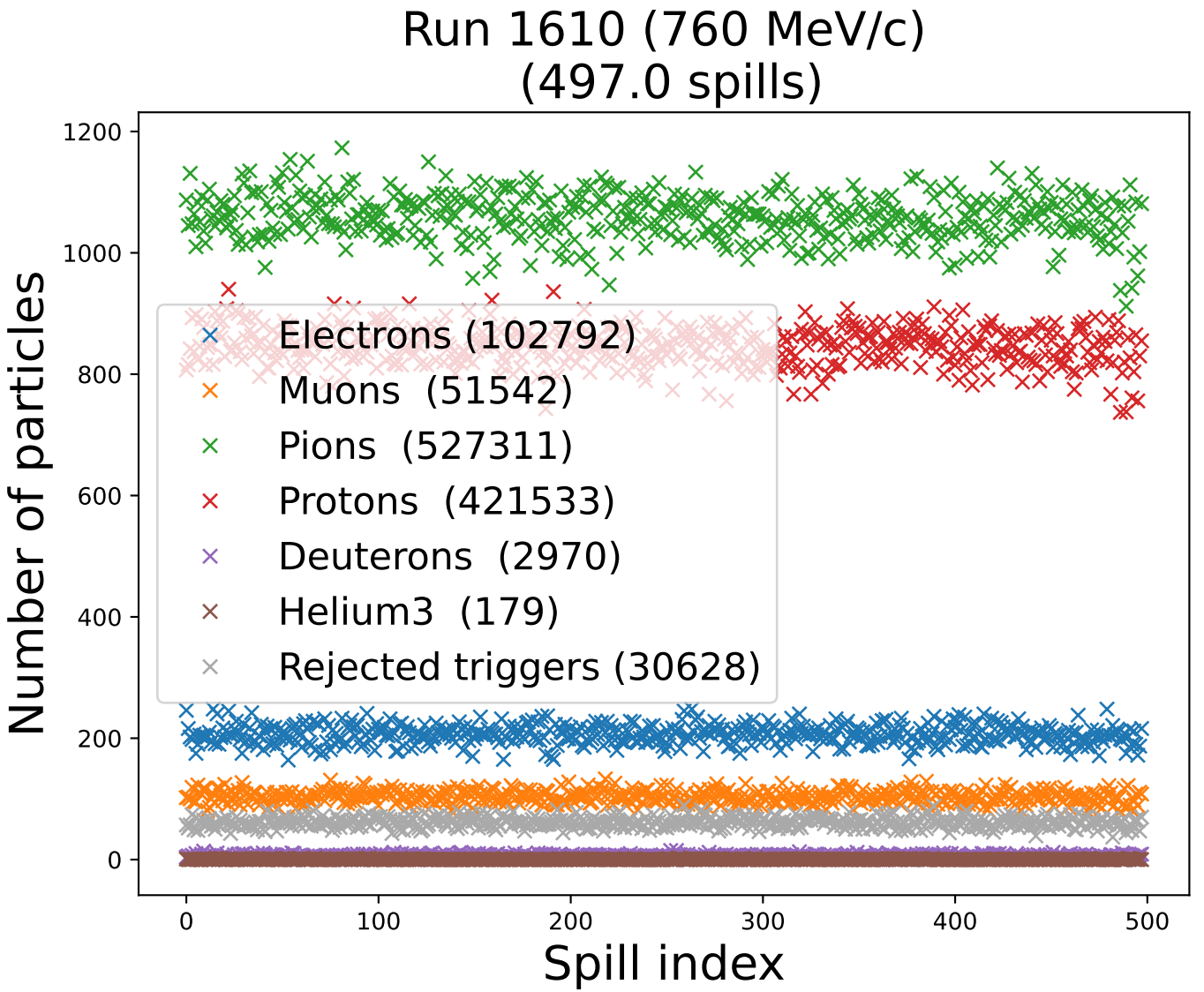

Run 1610 (760 MeV/c) - Muons
Predicted T0T1 TOF
Predicted T0T4 TOF
Measured T0T1 TOF:
14.80 +/- 8.9e-04 ns
Measured T0T4 TOF:
10.13 +/- 1.1e-03 ns
T0T1 TOF-est. momentum
Initial: 670.4 +/- 1.5 MeV/c
Final: 653.3 +/- 1.5 MeV/c
T0T4 TOF-est. momentum
Initial: 673.0 +/- 3.3 MeV/c
Final: 656.0 +/- 3.3 MeV/c
TOF (ns)
Initial momentum (MeV/c)
10
11
12
13
14
15
16
17
250
500
750
1000
1250
1500
1750

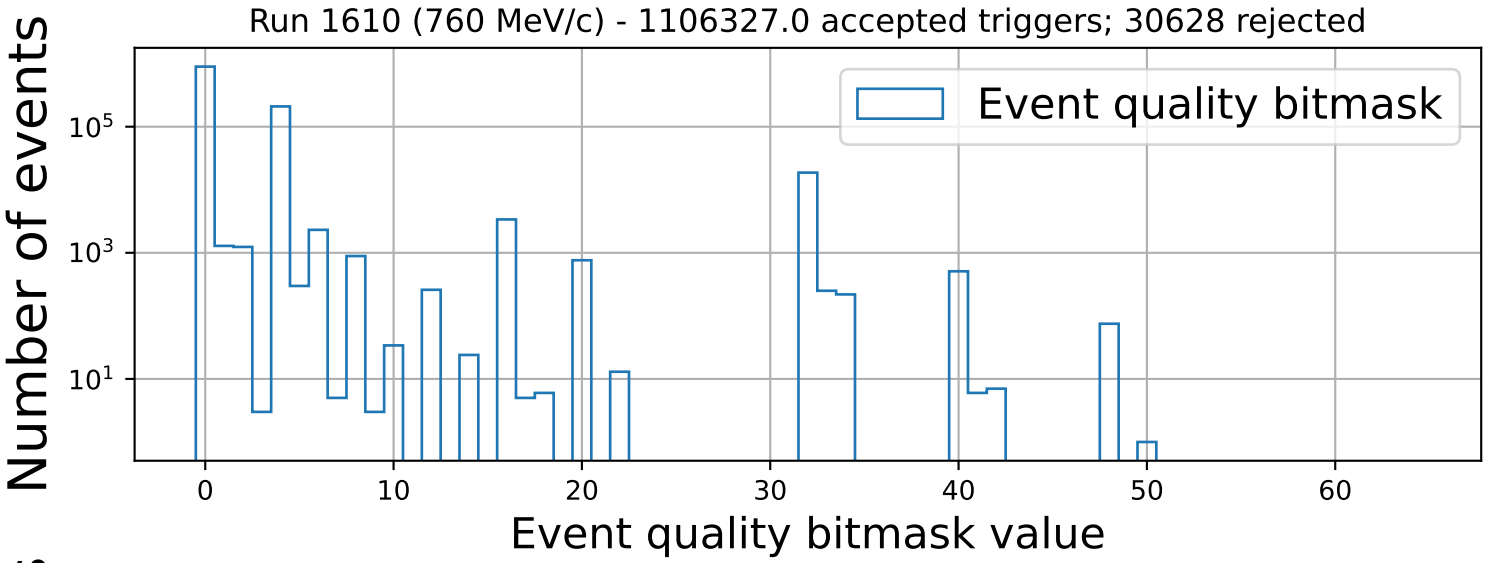

Run 1610 (760 MeV/c) - Pions
Predicted T0T1 TOF
Predicted T0T4 TOF
Measured T0T1 TOF:
14.87 +/- 2.5e-04 ns
Measured T0T4 TOF:
10.17 +/- 3.5e-04 ns
T0T1 TOF-est. momentum
Initial: 753.0 +/- 0.4 MeV/c
Final: 735.9 +/- 0.4 MeV/c
T0T4 TOF-est. momentum
Initial: 768.8 +/- 0.9 MeV/c
Final: 751.6 +/- 0.9 MeV/c
TOF (ns)
10
12
14
16
18
250
500
750
1000
1250
1500
1750
Initial momentum (MeV/c)

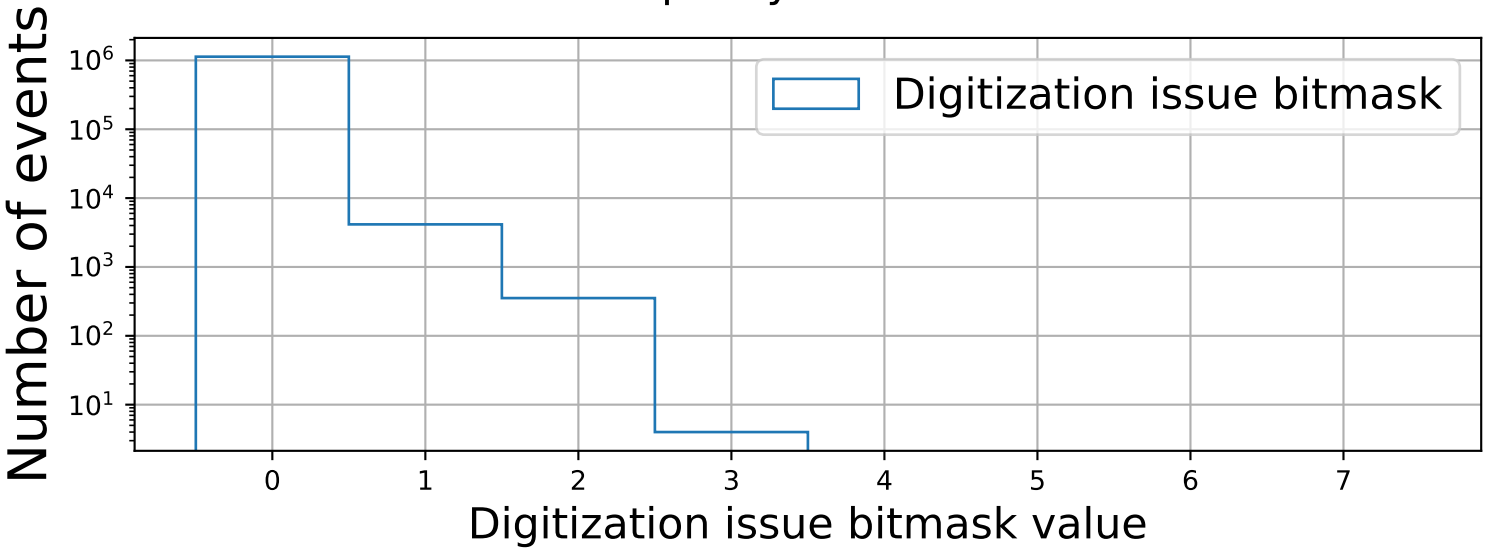

Run 1610 (760 MeV/c) - Protons
Predicted T0T1 TOF
Predicted T0T4 TOF
Measured T0T1 TOF:
23.69 +/- 5.2e-04 ns
Measured T0T4 TOF:
19.67 +/- 4.5e-04 ns
T0T1 TOF-est. momentum
Initial: 747.1 +/- 0.0 MeV/c
Final: 708.5 +/- 0.0 MeV/c
T0T4 TOF-est. momentum
Initial: 564.9 +/- 0.0 MeV/c
Final: 507.9 +/- 0.0 MeV/c
TOF (ns)
10
20
30
40
50
400
600
800
1000
1200
1400
1600
1800
Initial momentum (MeV/c)


Run 1610 (760 MeV/c) - Deuteron
Predicted T0T1 TOF
Predicted T0T4 TOF
Measured T0T1 TOF:
41.18 +/- 1.8e-02 ns
Measured T0T4 TOF:
29.37 +/- 4.5e-02 ns
T0T1 TOF-est. momentum
Initial: 741.8 +/- 0.3 MeV/c
Final: 629.4 +/- 0.4 MeV/c
T0T4 TOF-est. momentum
Initial: 700.1 +/- 1.1 MeV/c
Final: 574.1 +/- 1.5 MeV/c
TOF (ns)
45
40
35
30
25
20
15
600
800
1000
1200
1400
1600
1800
Initial momentum (MeV/c)


Run 1610 (760 MeV/c) - Helium3
Predicted T0T1 TOF
Predicted T0T4 TOF
Measured T0T1 TOF:
32.41 +/- 5.5e-02 ns
Measured T0T4 TOF:
32.41 +/- 5.5e-02 ns
T0T1 TOF-est. momentum
Initial: nan +/- nan MeV/c
Final: nan +/- nan MeV/c
T0T4 TOF-est. momentum
Initial: nan +/- nan MeV/c
Final: nan +/- nan MeV/c
TOF (ns)
37.5
35.0
32.5
30.0
27.5
25.0
22.5
20.0
1300
1400
1500
1600
1700
1800
1900
Initial momentum (MeV/c)


Run 1610 (760 MeV/c)
(497.0 spills)
Number of particles
Spill index
Electrons (102792)
Muons (51542)
Pions (527311)
Protons (421533)
Deuterons (2970)
Helium3 (179)
Rejected triggers (30628)
0
200
400
600
800
1000
1200
0
100
200
300
400
500


Run 1610 (760 MeV/c) - 1106327.0 accepted triggers; 30628 rejected
Event quality bitmask
Number of events
10^5
10^3
10^1
0
10
20
30
40
50
60
Event quality bitmask value


Digitization issue bitmask
Number of events
10^6
10^5
10^4
10^3
10^2
10^1
0
1
2
3
4
5
6
7
Digitization issue bitmask value